

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

Chap. 4 Open-Loop Discrete-Time Systems

Chap. 5 Closed-Loop Systems

Chap. 6 System Time-Response Characteristics

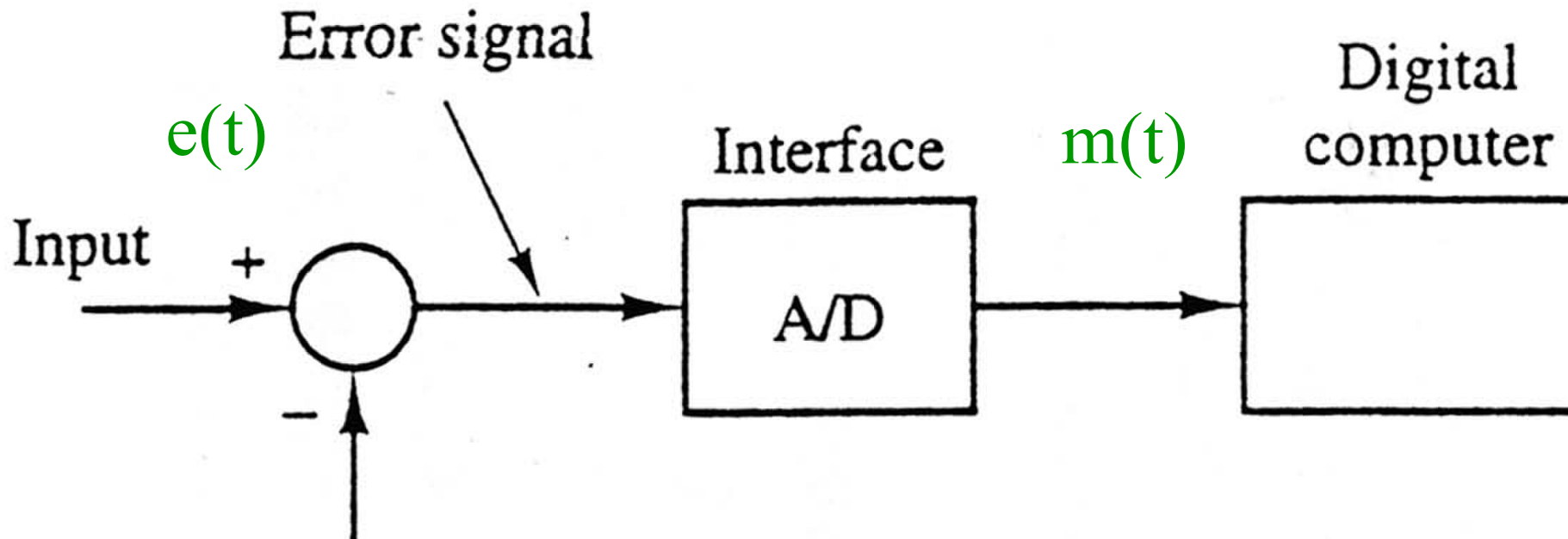
Chap. 7 Stability Analysis Techniques

Chap. 8 Digital Controller Design

2.2 Discrete-Time Systems

CT models: Differential Eq., Laplace transform, state space, diagrams

DT models: Difference Eq., z-transform, state space, diagrams



- Assume a PI controller

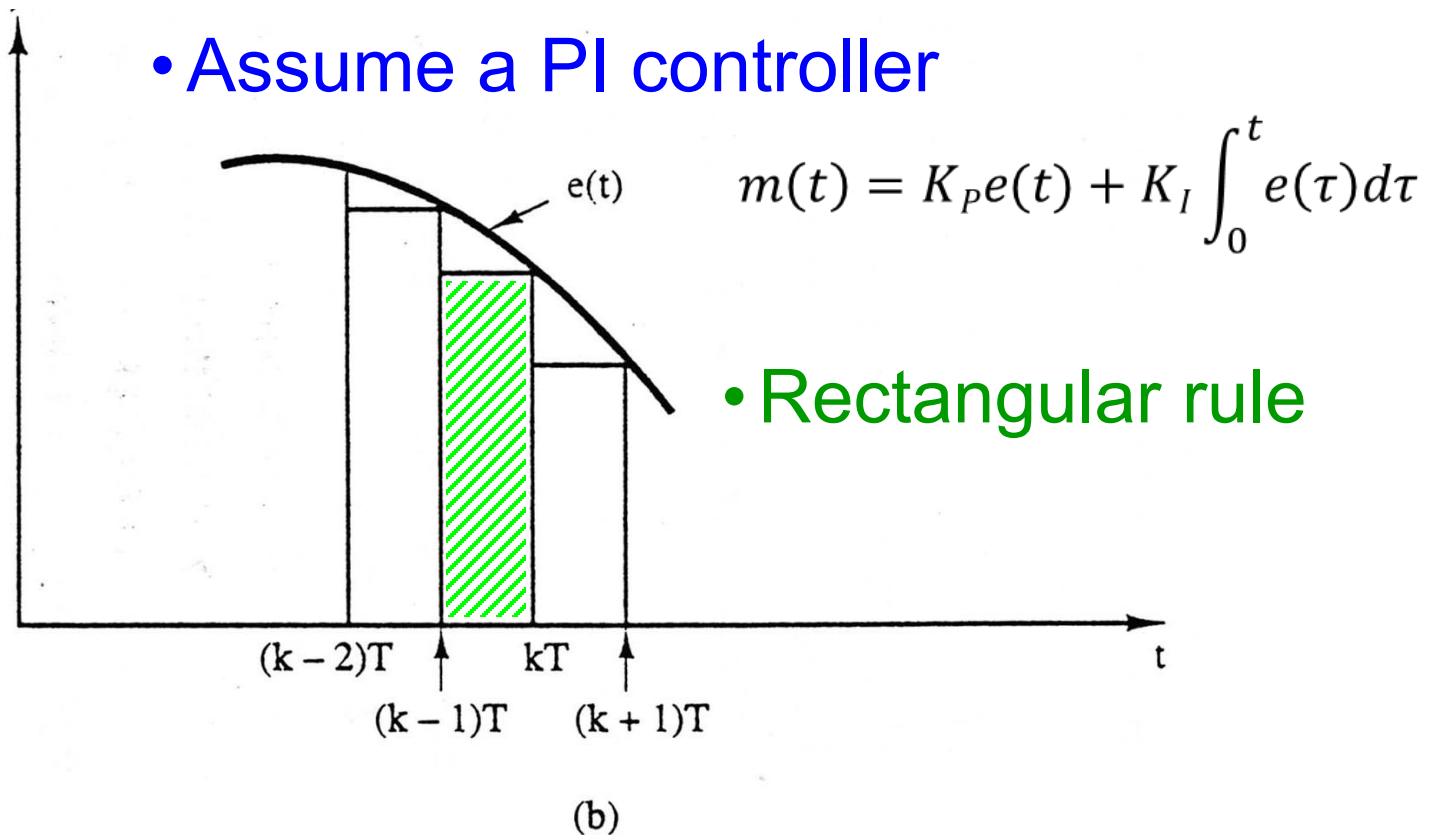


Figure 2-1 Digital control system.

- Two approaches
 - Analog design -> transform to digital (Ch11)
 - Digital design (Ch3-Ch10)

Realization

- Analog filter
 - RC network with OPA
- Digital filter (controller):
 - 1) T : sampling period
 - 2) n : order
 - 3) a_i and b_i : filter coefficients
- Round-off (truncation) issue! (四捨五入)
- Noise filtering required: differentiation in PID.
 - PD: high-pass filter

2.3 Transform Methods

The **z-transform** is expressed by the transform pair

$$\begin{aligned} E(z) &= \mathcal{Z}[\{e(k)\}] = e(0) + e(1)z^{-1} + e(2)z^{-2} + \dots \\ e(k) &= \mathcal{Z}^{-1}[E(z)] = \frac{1}{2\pi j} \oint_{\Gamma} E(z) z^{k-1} dz, \quad j = \sqrt{-1} \end{aligned} \quad (2-7)$$

$E(z)$ in (2-7) can be written in more compact notation as

$$E(z) = \mathcal{Z}[\{e(k)\}] = \sum_{k=0}^{\infty} e(k) z^{-k} \quad (2.8)$$

Applied to a sequence!

The **double-sided z-transform**, sometimes called the generating function is defined as

生成函數

一種形式冪級數，其每一項的係數
可以提供關於這個序列的信息。

$$G[\{f(k)\}] = \sum_{k=-\infty}^{\infty} f(k) z^{-k}$$

(2.9)

Example 2.1

反轉換 (inverse z-transform)

Given $E(z)$ below, find $\{e(k)\}$.

$$E(z) = 1 + 3z^{-1} - 2z^{-2} + z^{-4} + \dots$$

We know, then, from (2-7), that the values of the number sequence $\{e(k)\}$ are

$$e(0) = 1 \qquad e(3) = 0$$

$$e(1) = 3 \qquad e(4) = 1$$

$$e(2) = -2 \qquad \dots$$

Consider now the identity

等比級數

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots, \quad |x| < 1 \quad (2-10)$$

This power series is useful, in some cases, in expressing $E(z)$ in closed form, as will be illustrated in the next two examples.

封閉型式

Example 2.2

Given that $e(k) = 1$ for all k , find $E(z)$. By definition $E(z)$ is

$$E(z) = 1 + z^{-1} + z^{-2} + \dots$$

The closed form of $E(z)$ is obtained from (2-10):

$$E(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}, \quad |z^{-1}| < 1 \quad (2-11)$$

需標示，勿省略

Example 2.3

Given that $e(k) = \epsilon^{-akT}$, find $E(z)$. $E(z)$ can be written in power series form as

$$\begin{aligned} E(z) &= 1 + \epsilon^{-aT} z^{-1} + \epsilon^{-2aT} z^{-2} + \dots \\ &= 1 + (\epsilon^{-aT} z^{-1}) + (\epsilon^{-aT} z^{-1})^2 + \dots \end{aligned}$$

$E(z)$ can be put in closed form by applying (2-10), so that

$$E(z) = \frac{1}{1 - \epsilon^{-aT} z^{-1}} = \frac{z}{z - \epsilon^{-aT}}, \quad |\epsilon^{-aT} z^{-1}| < 1$$

Note that, in this example, $\{e(k)\}$ may be generated by sampling the function $e(t) = \epsilon^{-at}$.

取樣而得 (生成)

2.4 Properties of the z-Transform

1)

Addition and Subtraction

Linearity

Property. The z-transform of a sum of number sequences is equal to the sum of the z-transforms of the number sequences; that is,

$$\mathcal{Z}[e_1(k) \pm e_2(k)] = E_1(z) \pm E_2(z) \quad (2-12)$$

2)

Multiplication by a Constant

Property. The z-transform of a number sequence multiplied by a constant is equal to the constant multiplied by the z-transform of the number sequence:

$$\mathcal{Z}[ae(k)] = a\mathcal{Z}[e(k)] = aE(z) \quad (2-13)$$

Homogeneity 同質性

Example 2.4

Example 2.4

The linearity property of the z-transform can also be proved as follows. Let

$$e(k) = ae_1(k) + be_2(k)$$

Then

$$\begin{aligned}\mathcal{Z}[e(k)] &= \mathcal{Z}[ae_1(k) + be_2(k)] \\ &= \sum_{k=0}^{\infty} [ae_1(k) + be_2(k)]z^{-k} \\ &= a \sum_{k=0}^{\infty} e_1(k)z^{-k} + b \sum_{k=0}^{\infty} e_2(k)z^{-k} \\ &= a\mathcal{Z}[e_1(k)] + b\mathcal{Z}[e_2(k)] \\ &= aE_1(z) + bE_2(z)\end{aligned}$$

(START) Add Slides: Laplace's Transform.

Pierre-Simon, marquis de Laplace

(23 March 1749 – 5 March 1827)

Newton of France



LAPLACE TRANSFORM REVIEW

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt \quad (2.1)$$

$$\mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds = f(t)u(t) \quad (2.2)$$

where

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

z-Transform

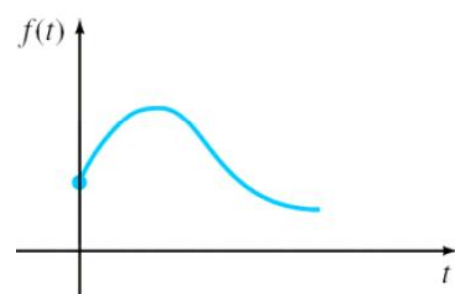
$$E(z) = \mathcal{Z}[\{e(k)\}] = \sum_{k=0}^{\infty} e(k)z^{-k}$$

$$\begin{aligned} E(z) &= \mathcal{Z}[\{e(k)\}] = e(0) + e(1)z^{-1} + e(2)z^{-2} + \dots \\ e(k) &= \mathcal{Z}^{-1}[E(z)] = \frac{1}{2\pi j} \oint_{\Gamma} E(z)z^{k-1} dz, \quad j = \sqrt{-1} \end{aligned} \quad (2-7)$$

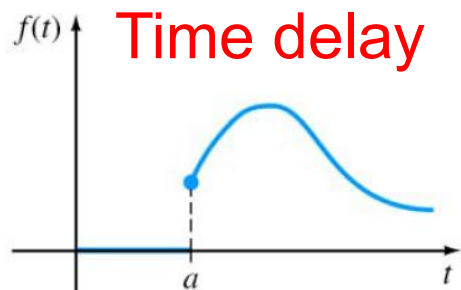
Quick Review

Time Shift: 2nd Translation Thm.

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$



(a) $f(t), t \geq 0$



(b) $f(t-a)u(t-a)$

Frequency Shift: 1st Translation Thm.

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

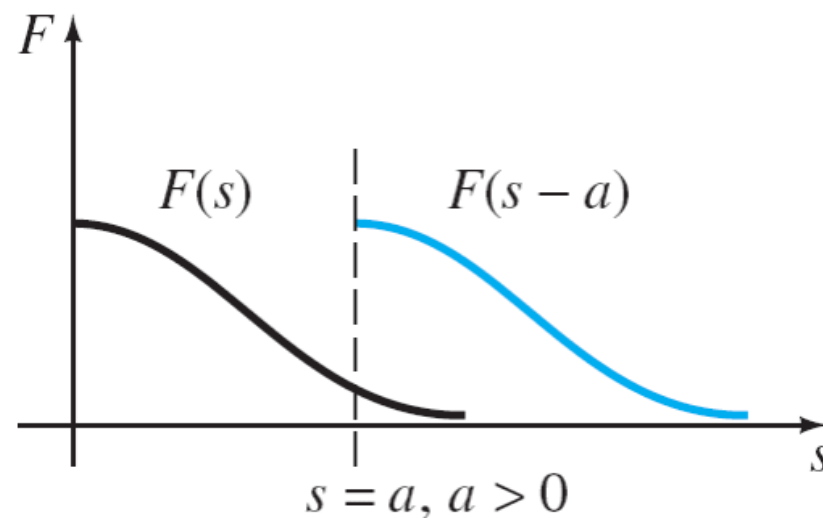


TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

TABLE 2.2 Laplace transform theorems

Item no.	Theorem		Name
1.	$\mathcal{L}[f(t)] = F(s)$	$= \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)]$	$= kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)]$	$= F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at} f(t)]$	$= F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)]$	$= e^{-sT} F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)]$	$= \frac{1}{a} F\left(\frac{s}{a}\right)$	Scaling theorem
7.	 $\mathcal{L}\left[\frac{df}{dt}\right]$	$= sF(s) - f(0-)$	Differentiation theorem
8.	 $\mathcal{L}\left[\frac{d^2f}{dt^2}\right]$	$= s^2F(s) - sf(0-) - \overset{\text{red dot}}{f'(0-)}$	Differentiation theorem
9.	 $\mathcal{L}\left[\frac{d^nf}{dt^n}\right]$	$= s^n F(s) - \sum_{k=1}^n s^{n-k} f^{k-1}(0-)$	Differentiation theorem
10.	 $\mathcal{L}\left[\int_{0-}^1 f(\tau) d\tau\right]$	$= \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty)$	$= \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+)$	$= \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

3) Real Translation Cf. Laplace's Time Shift

Property. Let n be a positive integer, and let $E(z)$ be the z -transform of $\{e(k)\}$. Then

$$\mathcal{Z}[e(k-n)u(k-n)] = \underbrace{z^{-n}}_{\text{Time delay}} E(z) \quad \text{Initial conditions} \quad (2-14)$$

and

$$\mathcal{Z}[e(k+n)u(k)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(k)z^{-k} \right] \quad (2-15)$$

Lose values

→ Jump to the next two slides.

4) Complex Translation Cf. Laplace's Frequency Shift

Property. Given that the z -transform of $e(k)$ is $E(z)$. Then

$$\mathcal{Z}[\epsilon^{ak} e(k)] = \underbrace{E(z\epsilon^{-a})}_{\text{Complex Translation}} \quad (2-17)$$

Real Translation (Time Shift)

Eq. (2-15): $n=1$, lose one values: $e(0)$

$n=2$, lose two values: $e(0)$, $e(1)$

TABLE 2-1 EXAMPLES OF SHIFTING

k	$e(k)$	$e(k - 2)$	$e(k + 2)$
0	2	0	1.3
1	1.6	0	1.1
2	1.3	2	1.0
3	1.1	1.6	...
4	1.0	1.3	...
...	...	...	...

Example 2.5

Example 2.5

It was shown in Example 2.3 that

Time delay

$$\mathcal{Z}[\epsilon^{-akT}] = \frac{z}{z - \epsilon^{-aT}}$$

Thus

$$\mathcal{Z}[\epsilon^{-a(k-3)T} u[(k-3)T]] = z^{-3} \left[\frac{z}{z - \epsilon^{-aT}} \right] = \frac{1}{z^2(z - \epsilon^{-aT})} \rightarrow \text{No loss}$$

where $u(kT)$ is the unit step. Also,

$$\mathcal{Z}[\epsilon^{-a(k+2)T} u(kT)] = z^2 \left[\frac{z}{z - \epsilon^{-aT}} - 1 - \epsilon^{-aT} z^{-1} \right] \rightarrow \text{Lose two values}$$

We now define the discrete unit impulse function $\delta(k - n)$:

$$\delta(k - n) = \begin{cases} 1, & k = n \\ 0, & k \neq n \end{cases} \quad (2-16)$$

This function is also called the unit sample function. The z-transform of the unit impulse function is then, for $n \geq 0$,

$$\mathcal{Z}[\delta(k - n)] = \sum_{k=0}^{\infty} \delta(k - n) z^{-k} = \delta(k - n) z^{-n} = z^{-n}$$

from (2-16).

Add Slides: Dirac Delta Function.

(START) Add Slides: Dirac Delta Function.

7.5 The Dirac Delta Function

Paul Adrian Maurice Dirac (1902-1984)



Received the Nobel Prize in Physics, 1933.

原子理論

“for the discovery of new productive forms of atomic theory.”

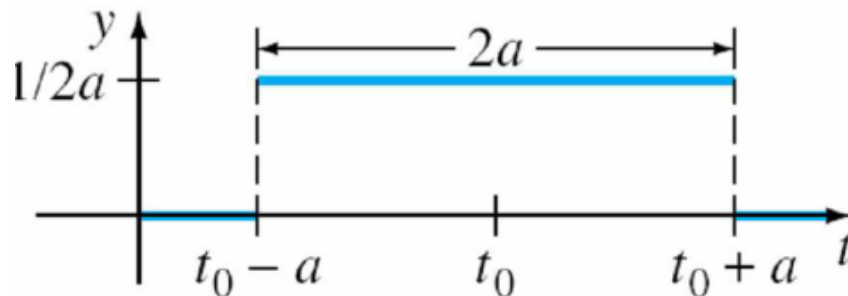
Unit Impulse

The function

$$\delta_a(t - t_0) = \begin{cases} 0, & 0 \leq t < t_0 - a \\ \frac{1}{2a}, & t_0 - a \leq t < t_0 + a \\ 0, & t \geq t_0 + a \end{cases}$$

is called a unit impulse, because it possesses the integration property:

$$\int_0^{\infty} \delta_a(t - t_0) dt = 1$$



(a) graph of $\delta_a(t - t_0)$

Fig. 7.5.2

Dirac Delta Function

A “function” that approximates $\delta_a(t - t_0)$ and is defined by the limit

$$\delta(t - t_0) = \lim_{a \rightarrow 0} \delta_a(t - t_0)$$

- 2 properties:

$$(i) \quad \delta(t - t_0) = \begin{cases} \infty, & t = t_0 \\ 0, & t \neq t_0 \end{cases}$$

$$(ii) \quad \int_0^{\infty} \delta(t - t_0) dt = 1$$

integration property

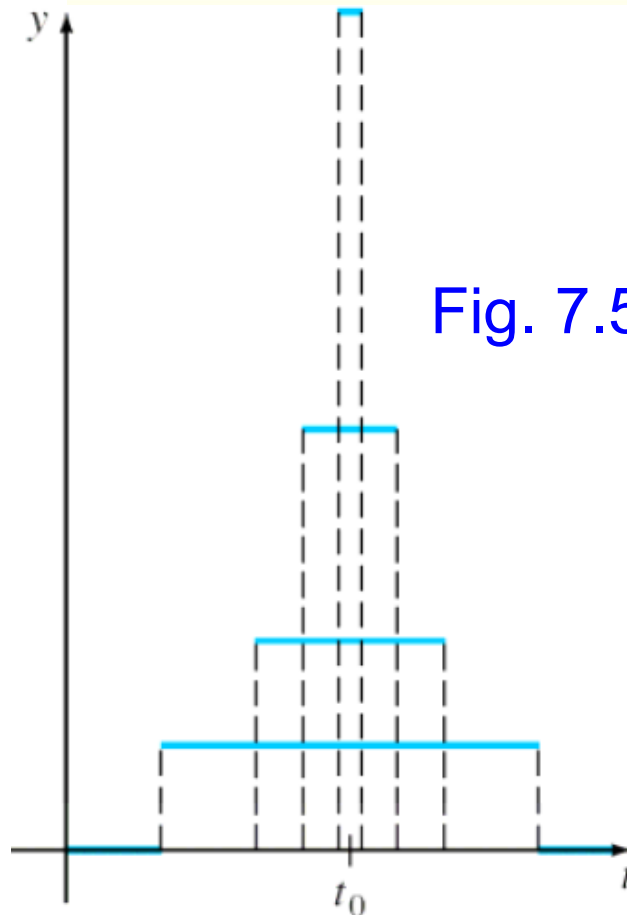


Fig. 7.5.2

(b) behavior of δ_a as $a \rightarrow 0$

Recall from Thm. 7.1.3: $\lim_{s \rightarrow \infty} F(s) = 0$
(P260, slide 13)

Thm 7.5.1 Transform of the Dirac Delta Function

For $t_0 > 0$,

Time delay $\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}$

Furthermore,

$$\mathcal{L}\{\delta(t)\} = 1 \quad (\text{as } t_0=0)$$

Recall from 7.3.2 (P275, slide 35)

$$\mathcal{L}\{u(t - a)\} = \frac{e^{-as}}{s}$$

- Ex-1: Two IVP Problems (Spring-Mass Systems)

$$y'' + y = 4\delta(t - 2\pi)$$

$$(a) \ y(0) = 1, \ y'(0) = 0$$

$$(b) \ y(0) = 0, \ y'(0) = 0$$

- See Remarks

- Properties of Delta Functions

Sampling:
(definition)

$$\int_0^{\infty} \delta(t - t_0) \underline{f(t)} dt = f(t_0)$$

$$\begin{aligned} f(t) &= 1 \\ f(t) &= e^{-st} \end{aligned}$$

Integration
property

Translation:

$$f(t) * \delta(t - t_0) = f(t - t_0)$$

s-Transform

- | | | | |
|-----|-------------|---------------------------------------|------------------------------------|
| 11. | $f(\infty)$ | $= \lim_{s \rightarrow 0} sF(s)$ | Final value theorem ¹ |
| 12. | $f(0+)$ | $= \lim_{s \rightarrow \infty} sF(s)$ | Initial value theorem ² |
-

5)

Initial Value

Property. Given that the z -transform of $e(k)$ is $E(z)$. Then

$$e(0) = \lim_{z \rightarrow \infty} E(z) \quad (2-18)$$

Proof. Since

$$E(z) = e(0) + e(1)z^{-1} + e(2)z^{-2} + \dots$$

then (2-18) is seen by inspection.

6)

Final Value → 求穩態值 → steady-state error

Property. Given that the z -transform of $e(k)$ is $E(z)$. Then

$$\lim_{n \rightarrow \infty} e(n) = \lim_{z \rightarrow 1} (z - 1)E(z)$$

provided that the left-side limit exists.

Example 2.7

Example 2.7

To illustrate the initial-value property and the final-value property, consider the z -transform of $e(k) = 1, k = 0, 1, 2, \dots$. We have shown, in Example 2.2, that

$$\underline{E(z) = \mathfrak{z}[1] = \frac{z}{z-1}}$$

Applying the initial-value property, we see that

$$e(0) = \lim_{z \rightarrow \infty} \frac{z}{z-1} = \lim_{z \rightarrow \infty} \frac{1}{1-1/z} (= 1)$$

Since the final value of $e(k)$ exists, we may apply the final-value property.

$$\lim_{k \rightarrow \infty} e(k) = \lim_{z \rightarrow 1} (z-1)E(z) = \lim_{z \rightarrow 1} z (= 1)$$

TABLE 2-2 PROPERTIES OF THE z -TRANSFORM

		Sequence	Transform
		$e(k)$	★ $E(z) = \sum_{k=0}^{\infty} e(k)z^{-k}$
Linearity		$a_1 e_1(k) + a_2 e_2(k)$	$a_1 E_1(z) + a_2 E_2(z)$
★ Real Trans.	{	$e(k - n)u(k - n); \quad n \geq 0$	😊 $z^{-n} E(z)$
		$e(k + n)u(k); \quad n \geq 1$	$z^n \left[E(z) - \sum_{k=0}^{n-1} e(k)z^{-k} \right]$
Complex Trans.	{	$\epsilon^{ak} e(k)$	$E(z\epsilon^{-a})$
		$ke(k)$	$-z \frac{dE(z)}{dz}$
Convolution	{	★ $e_1(k) * e_2(k)$	$E_1(z)E_2(z)$
		$e_1(k) = \sum_{n=0}^k e(n)$	$E_1(z) = \frac{z}{z-1} E(z)$
		Initial value: $e(0) = \lim_{z \rightarrow \infty} E(z)$	
		Final value: $e(\infty) = \lim_{z \rightarrow 1} (z-1)E(z),$	if $e(\infty)$ exists

2.5 Solution of Difference Equations

Three Approaches

1) Classical approach

- $m = m_c$ (complementary: homo.) + m_p (particular)

2) Sequential procedure

- Digital-computer solution

3) z-transform

By z-Transform

Consider the following n th-order difference equation, where it is assumed that $\{e(k)\}$ is known.

$$\begin{aligned} m(k) + a_{n-1}m(k-1) + \cdots + a_0m(k-n) \\ = b_n e(k) + b_{n-1}e(k-1) + \cdots + b_0e(k-n) \end{aligned} \quad (2-19)$$

The z-transform of (2-19), which results from the use of the real translation property (2-14), is

$$\begin{aligned} M(z) + a_{n-1}z^{-1}M(z) + \cdots + a_0z^{-n}M(z) \\ = b_n E(z) + b_{n-1}z^{-1}E(z) + \cdots + b_0z^{-n}E(z) \end{aligned} \quad (2-20)$$

Note that the z-transform has changed the difference equation in (2-19) to the algebraic equation in (2-20). Solving the expression above for $M(z)$ yields

$$M(z) = \frac{b_n + b_{n-1}z^{-1} + \cdots + b_0z^{-n}}{1 + a_{n-1}z^{-1} + \cdots + a_0z^{-n}} E(z) \quad (2-21)$$

$m(k)$ can be found by taking the inverse z-transform of (2-21). General techniques for determining the inverse z-transform are discussed in the next section.

Example 2.9

Example 2.9

Consider the difference equation of Example 2.8.

$$m(k) = e(k) - e(k-1) - m(k-1)$$

The z -transform of this equation, obtained via the real translation property, is

$$M(z) = E(z) - z^{-1}E(z) - z^{-1}M(z)$$

or

$$M(z) = \frac{z-1}{z+1}E(z)$$

Since

$$\underline{e(k) = \begin{cases} 1, & k \text{ even} \\ 0, & k \text{ odd} \end{cases}} \quad \text{From Example 2.8}$$

we see that

$$E(z) = 1 + z^{-2} + z^{-4} + \dots = \frac{1}{1 - z^{-2}} = \frac{z^2}{z^2 - 1} = \frac{z^2}{(z-1)(z+1)}$$

Thus

$$\underline{M(z) = \frac{z-1}{z+1} \frac{z^2}{(z-1)(z+1)} = \frac{z^2}{z^2 + 2z + 1}}$$

We can expand $M(z)$ into a power series by dividing the numerator of $M(z)$ by its

Power Series Method (Direct division)

denominator so as to obtain

$$\begin{array}{r}
 1 - 2z^{-1} + 3z^{-2} - 4z^{-3} + \dots \\
 z^2 + 2z + 1 \overline{) z^2} \\
 \underline{z^2 + 2z + 1} \\
 -2z - 1 \\
 \underline{-2z - 4 - 2z^{-1}} \\
 3 + 2z^{-1} \\
 \underline{3 + 6z^{-1} + 3z^{-2}} \\
 -4z^{-1} - 3z^{-2} \\
 \dots
 \end{array}$$

(We discuss this method for finding the inverse transform in the next section)
Therefore,

$$\begin{array}{l}
 m(k) \Rightarrow m(0)=1, m(1)=-2, \\
 m(2)=3, m(3)=-4\dots
 \end{array}
 \quad
 \underline{M(z) = 1 - 2z^{-1} + 3z^{-2} - 4z^{-3} + \dots}$$

- Solution in **a closed form**? Partial-Fraction Expansion Method. → **Next**
 ✓ **$m(k) = (-1)^k(1+k)$** (補充:p39)
- Initial conditions of difference equations: **$k = [0, n-1]$** (see Tab. 2-2)
 - Compare with *differential* equations: **$t = 0$**

2.6 The Inverse z-Transform

1) Power Series Method (Direct division)

- Solution in **an open form**.

$$E(z) = e_0 + e_1 z^{-1} + e_2 z^{-2} + \dots \quad (2-24)$$

2) Partial-Fraction Expansion Method ★

- Solution in **a closed form**.

3) Inversion Formula Method (Inverse integral)

- Solution in **a closed form**.

4) Discrete Convolution

- Solution in **an open form**.

Example 2.10

Example 2.10

It is desired to find the values of $e(k)$ for $E(z)$ given by the expression

$$E(z) = \frac{z}{z^2 - 3z + 2}$$

Using long division, we obtain

$$\begin{array}{r} z^{-1} + 3z^{-2} + 7z^{-3} + 15z^{-4} + \dots \\ z^2 - 3z + 2 \overline{)z} \\ \underline{z - 3 + 2z^{-1}} \\ 3 - 2z^{-1} \\ \underline{3 - 9z^{-1} + 6z^{-2}} \\ 7z^{-1} - 6z^{-2} \\ \underline{7z^{-1} - 21z^{-2} + 14z^{-3}} \\ 15z^{-2} - 14z^{-3} + \dots \\ \dots \end{array}$$

and therefore

$$e(0) = 0 \quad e(4) = 15$$

$$e(1) = 1 \quad \dots$$

$$e(2) = 3 \quad e(k) = 2^k - 1$$

$$e(3) = 7 \quad \dots$$

Not easy to
recognize the
general expression.

Partial-Fraction Expansion Method

TABLE 2-3 z-TRANSFORMS

Sequence		z-Transform	
$\delta(k - n)$	★	z^{-n}	Special case: $n=0$ for 常數項
1	★	$\frac{z}{z-1}$	Pole = 1
k		$\frac{z}{(z-1)^2}$	
k^2		$\frac{z(z+1)}{(z-1)^3}$	
a^k	★	$\frac{z}{z-a}$	Pole = a
ka^k		$\frac{az}{(z-a)^2}$	
$\sin ak$		$\frac{z \sin a}{z^2 - 2z \cos a + 1}$	Example 2.9
$\cos ak$		$\frac{z(z - \cos a)}{z^2 - 2z \cos a + 1}$	
$a^k \sin bk$		$\frac{az \sin b}{z^2 - 2az \cos b + a^2}$	
$a^k \cos bk$		$\frac{z^2 - az \cos b}{z^2 - 2az \cos b + a^2}$	

More: See Appendix VIII.

Example 2.11

Real (distinct) Poles

Consider the function $E(z)$ given in Example 2.10:

$$E(z) = \frac{z}{(z-1)(z-2)}$$

Hence

$$\frac{E(z)}{z} = \frac{1}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{1}{z-2}$$

Then

$$\mathcal{Z}^{-1}[E(z)] = \mathcal{Z}^{-1}\left[\frac{-z}{z-1}\right] + \mathcal{Z}^{-1}\left[\frac{z}{z-2}\right]$$

From (2-26) or Table 2-3, the value of $e(k)$ is given by

$$e(k) = -1 + 2^k$$

which is the same value as that found in Example 2.10.

Consider next the function

$$E_1(z) = z^{-1} E(z) = \frac{1}{(z-1)(z-2)}$$

From the real translation property (2-14), $e_1(k)$ is given by

方法-1

$$\begin{aligned} e_1(k) &= \mathcal{Z}^{-1}[z^{-1} E(z)] = e(k-1)u(k-1) \\ &= [-1 + 2^{(k-1)}]u(k-1) \\ &= \begin{cases} 0, & k = 0 \\ -1 + 2^{(k-1)}, & k \geq 1 \end{cases} \end{aligned}$$

1

① = ②

方法-2

This inverse can also be found by partial-fraction expansion.

$$\frac{E_1(z)}{z} = \frac{1}{z(z-1)(z-2)} = \frac{\frac{1}{2}}{z} + \frac{-1}{z-1} + \frac{\frac{1}{2}}{z-2}$$

Or,

$$E_1(z) = \frac{1}{2} + \frac{-z}{z-1} + \frac{(\frac{1}{2})z}{z-2}$$

Thus

$$e_1(k) = \underline{a} - 1 + (\frac{1}{2})(2)^k = a - 1 + (2)^{k-1}$$

where

$$a = \begin{cases} \frac{1}{2}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

since from Table 2-3,

Special case: n=0

$$\mathcal{Z}^{-1}\left[\frac{1}{2}\right] = \underline{\frac{1}{2}\delta(k)} = \begin{cases} \frac{1}{2}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

②

Complex Poles

solution

$$\underline{y(k) = A\epsilon^{akT} \cos(bkT + \theta) = \frac{A\epsilon^{akT}}{2} [\epsilon^{jbkT} \epsilon^{j\theta} + \epsilon^{-jbkT} \epsilon^{-j\theta}]} \quad (2-27)$$

$$= \frac{A}{2} [\epsilon^{(aT + jbT)k} \epsilon^{j\theta} + \epsilon^{(aT - jbT)k} \epsilon^{-j\theta}]$$

where a and b are real. Euler's relation, given by

$$\cos x = \frac{\epsilon^{jx} + \epsilon^{-jx}}{2}$$

is used in (2-27). The z-transform of this function is given by, from Appendix VIII,

$$\begin{aligned}
 Y(z) &= \frac{A}{2} \left[\frac{\epsilon^{j\theta} z}{z - \epsilon^{aT + jbT}} + \frac{\epsilon^{-j\theta} z}{z - \epsilon^{aT - jbT}} \right] \\
 &= \frac{(A\epsilon^{j\theta}/2)z}{z - \epsilon^{aT + jbT}} + \frac{(A\epsilon^{-j\theta}/2)z}{z - \epsilon^{aT - jbT}} = \frac{k_1 z}{z - p_1} + \frac{k_1^* z}{z - p_1^*}
 \end{aligned} \quad (2-28)$$

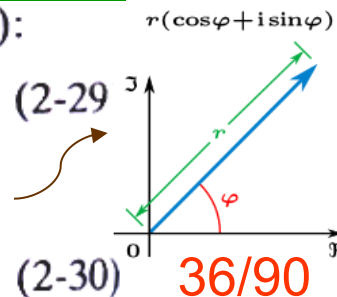
where the asterisk indicates the complex conjugate.

The usual partial-fraction expansion yields terms in the form of (2-28). Hence, given the partial-fraction coefficient k_1 and the pole p_1 in (2-28), we can solve for the discrete-time function of (2-27) using the following relationship from (2-28):

coeffi.
and

$$\left\{ \begin{aligned}
 p_1 &= \epsilon^{aT} \epsilon^{jbT} = \epsilon^{aT} \angle bT \Rightarrow aT = \ln|p_1|; & bT &= \arg p_1 \\
 k_1 &= \frac{A\epsilon^{j\theta}}{2} = \frac{A}{2} \angle \theta \Rightarrow A = 2|k_1|; & \theta &= \arg k_1
 \end{aligned} \right.$$

argument



Example 2.12

We find the inverse z-transform of the function

$$Y(z) = \frac{-3.894z}{z^2 + 0.6065} = \frac{-3.894z}{(z - j0.7788)(z + j0.7788)}$$
$$= \frac{k_1 z}{z - j0.7788} + \frac{k_1^* z}{z + j0.7788}$$

Dividing both sides by z , we calculate k_1 :

$$k_1 = (z - j0.7788) \left[\frac{-3.894}{(z - j0.7788)(z + j0.7788)} \right]_{z = j0.7788}$$
$$= \frac{-3.894}{z + j0.7788} \Big|_{z = j0.7788} = \frac{-3.894}{2(j0.7788)} = 2.5 \angle 90^\circ$$

From (2-29) and (2-30), with $p_1 = j0.7788$,

$$aT = \ln |p_1| = \ln(0.7788) = -0.250; \quad bT = \arg p_1 = \pi/2$$

$$A = 2|k_1| = 2(2.5) = 5; \quad \theta = \arg k_1 = \pi/2$$

Hence, from (2-27),

$$y(k) = A e^{akT} \cos(bkT + \theta)$$
$$= 5 e^{-0.25k} \cos\left(\frac{\pi}{2}k + \frac{\pi}{2}\right) = -5 e^{-0.25k} \sin \frac{\pi}{2}k$$

coeff. 

Inversion-Formula Method 反轉公式

Perhaps the most general technique for obtaining the inverse of a z -transform is the inversion integral. This integral, derived via complex variable theory, is, from (2-7),

$$e(k) = \frac{1}{2\pi j} \oint_{\Gamma} E(z) z^{k-1} dz, \quad j = \sqrt{-1} \quad (2-31)$$

This expression is the line integral in the z -plane along the closed path Γ , where Γ is any path that encloses all the finite poles of $E(z)z^{k-1}$ [5,7].

Using the theorem of residues [7], we can evaluate the integral in (2-31) via the expression

$$e(k) = \sum_{\substack{\text{at poles} \\ \text{of } [E(z)z^{k-1}]}} [\text{residues of } E(z)z^{k-1}] \quad (2-32)$$

If the function $E(z)z^{k-1}$ has a simple pole at $z = a$, the residue is evaluated

$$(\text{residue})_{z=a} = (z - a)E(z)z^{k-1} \Big|_{z=a} \quad (2.33)$$

For a pole of order m at $z = a$, the residue is calculated using the expression

$$(\text{residue})_{z=a} = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-a)^m E(z)z^{k-1}] \Big|_{z=a} \quad (2.34)$$

Example 2.13

Consider the function $E(z)$ from Examples 2.10 and 2.11:

$$E(z) = \frac{z}{(z-1)(z-2)}$$

Substituting this expression into (2-32) and (2-33) yields

$$e(k) = \frac{z^k}{z-2} \Big|_{z=1} + \frac{z^k}{z-1} \Big|_{z=2} = \underline{-1 + 2^k}$$

and the result is seen to be the same as that obtained in the previous examples. As
Example 2.11, let

$$E_1(z) = z^{-1} E(z) = \frac{1}{(z-1)(z-2)}$$

Then, from the real translation property,

$$e_1(k) = e(k-1)u(k-1) = \underline{[-1 + 2^{(k-1)}]u(k-1)}$$

We can also find $e_1(k)$ by the inversion formula. In (2-32),

$$E_1(z)z^{k-1} = \frac{z^{k-1}}{(z-1)(z-2)}$$

This function has a pole at $z = 0$ only for $k = 0$, and thus

$$\begin{aligned} e_1(0) &= \sum_{\text{at } z=0,1,2} \left[\text{residues of } \frac{1}{z(z-1)(z-2)} \right] \\ &= \frac{1}{2} - 1 + \frac{1}{2} = 0 \end{aligned}$$

...left as an exercise.

Discrete Convolution 離散迴旋

The discrete convolution technique used for determining the inverse z -transform is analogous to the convolution integral employed in the use of Laplace transforms. Suppose that the function $E(z)$ can be expressed as the product of two functions, each of which, in general, will be simpler than $E(z)$; that is,

$$E(z) = E_1(z)E_2(z) \quad (2-35)$$

Further, let $E_1(z)$ and $E_2(z)$ be expressed as power series. Then

$$E(z) = [e_1(0) + e_1(1)z^{-1} + e_1(2)z^{-2} + \cdots][e_2(0) + e_2(1)z^{-1} + e_2(2)z^{-2} + \cdots] \quad (2-36)$$

Direct multiplication of the two power series yields

$$E(z) = e_1(0)e_2(0) + [e_1(0)e_2(1) + e_1(1)e_2(0)]z^{-1} + [e_1(0)e_2(2) + e_1(1)e_2(1) + e_1(2)e_2(0)]z^{-2} + \cdots \quad (2-37)$$

Thus the general relationship for $e(k)$ is seen to be

$$\begin{aligned} e(k) &= e_1(0)e_2(k) + e_1(1)e_2(k-1) + \cdots + e_1(k)e_2(0) \\ &= \sum_{n=0}^k e_1(n)e_2(k-n) = \sum_{n=0}^k e_1(k-n)e_2(n) \end{aligned} \quad (2-38)$$

Equation (2-38) is the discrete convolution summation, and may be useful in determining the inverse z -transform of a function $E(z) = E_1(z)E_2(z)$ if $E_1(z)$ and $E_2(z)$ are initially expressed as power series. Convolution is usually denoted as

$$e(k) = \mathcal{Z}^{-1}[E_1(z)E_2(z)] = e_1(k) * e_2(k) \quad (2-39)$$

Example 2.15

Once again consider the function $E(z)$ from Example 2.10:

$$E(z) = \frac{z}{(z-1)(z-2)} = E_1(z)E_2(z)$$

where we define

$$E_1(z) = \frac{z}{z-1} = 1 + z^{-1} + z^{-2} + \dots$$

and

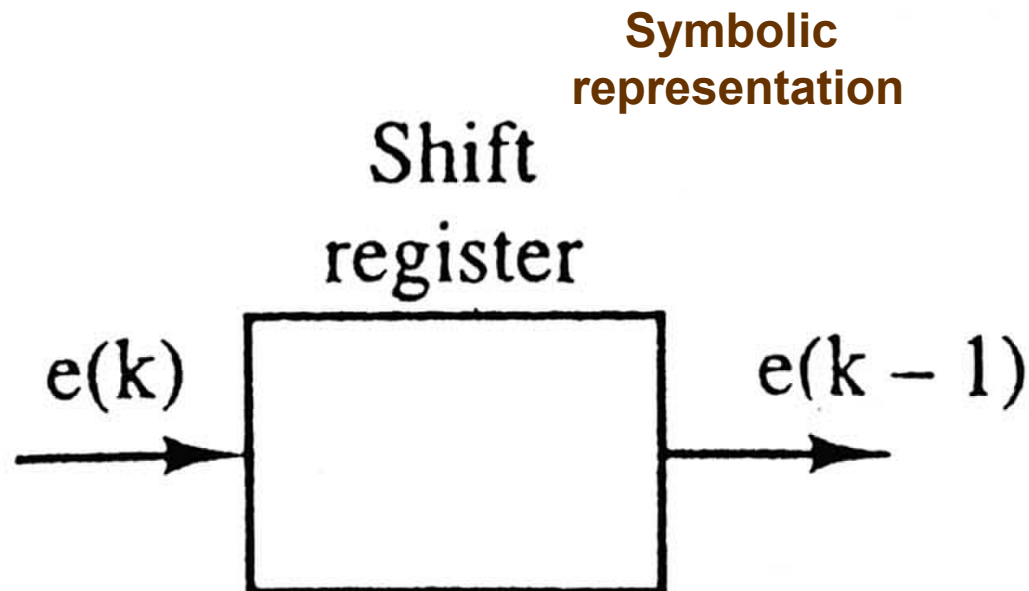
$$E_2(z) = \frac{1}{z-2} = z^{-1} + 2z^{-2} + (2)^2 z^{-3} + \dots$$

These expansions are obtained by the power-series method. Then $e(k)$ can be formed directly from (2-38). For example, $e(3)$ is given by the expression

$$\begin{aligned} e(3) &= \sum_{n=0}^3 e_1(n)e_2(3-n) \\ &= e_1(0)e_2(3) + e_1(1)e_2(2) + e_1(2)e_2(1) + e_1(3)e_2(0) \\ &= (1)(2^2) + (1)(2) + (1)(1) + (1)(0) = 7 \end{aligned}$$

2.7 Simulation Diagrams & Flow Graphs

- **System Modeling**
 - Diff. Equations
 - Transfer Functions
 - State Space
 - **Diagrams**: Block diagrams, Signal flow charts

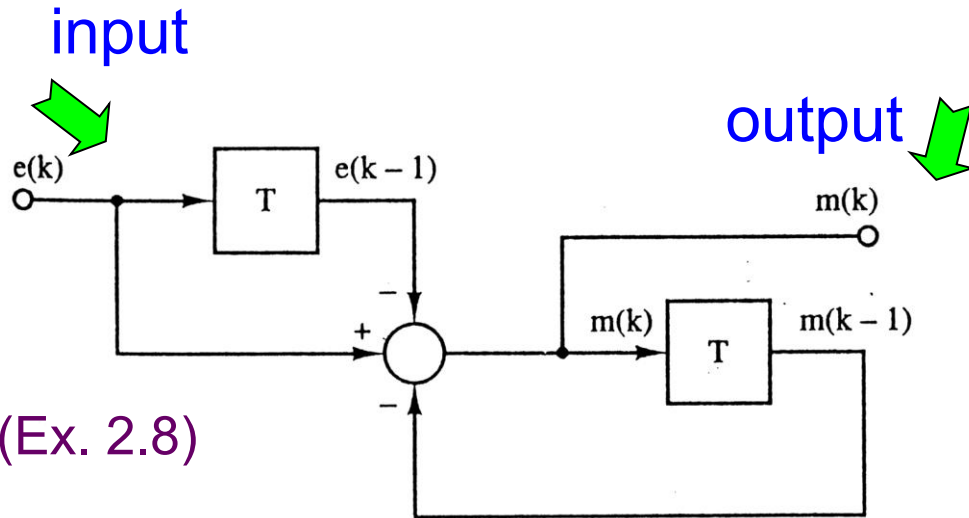


1st-order DE

$$m(k) = e(k) - e(k-1) - m(k-1)$$

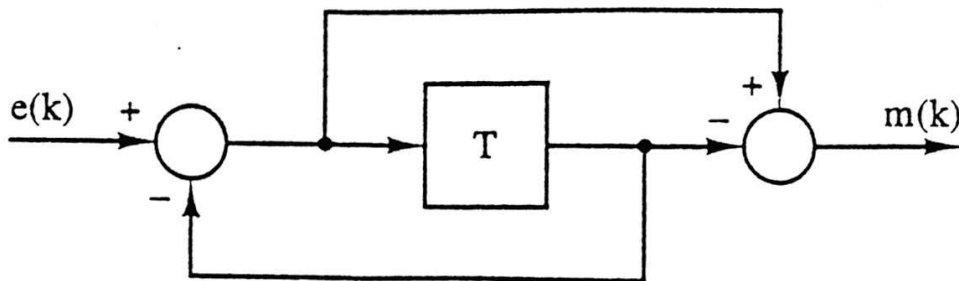
(2.40)

Figure 2-3 Simulation diagram for the difference equation used in Example 2.8.



Special-purpose vs.
general purpose computers (Ex. 2.8)

Minimal representation (n delays only)

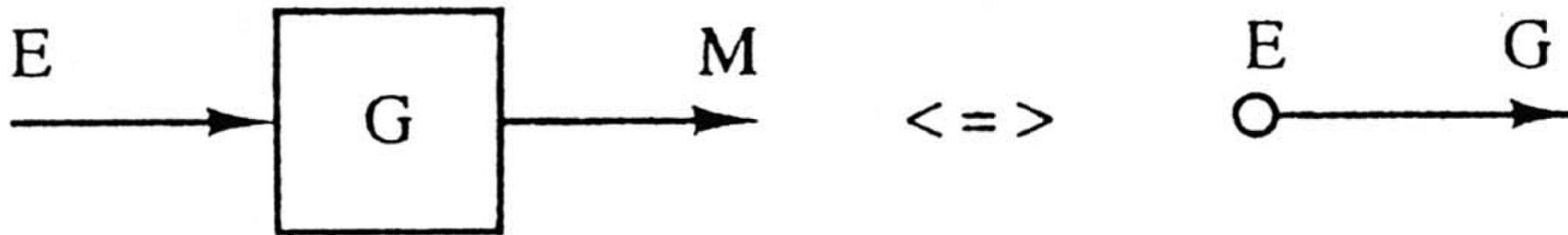


This representation may be verified
via Mason's gain formula.

Figure 2-7 A different simulation diagram for Example 2.8.

Basic Elements

- Continuous systems: integrator.
- Discrete systems: **time delay** (memory) of T sec.



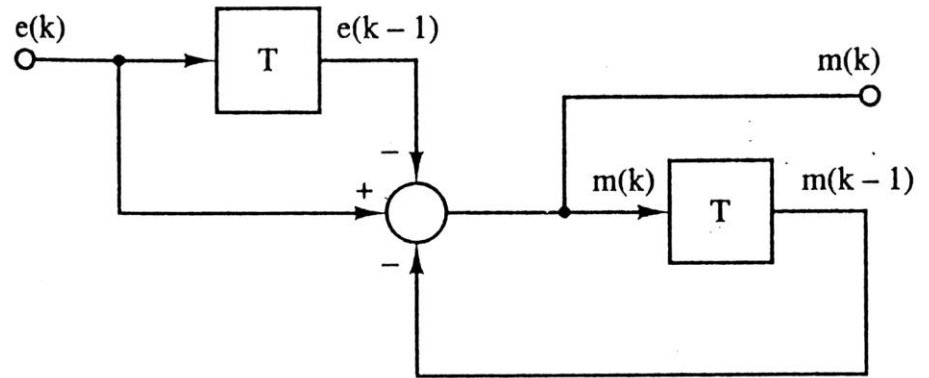
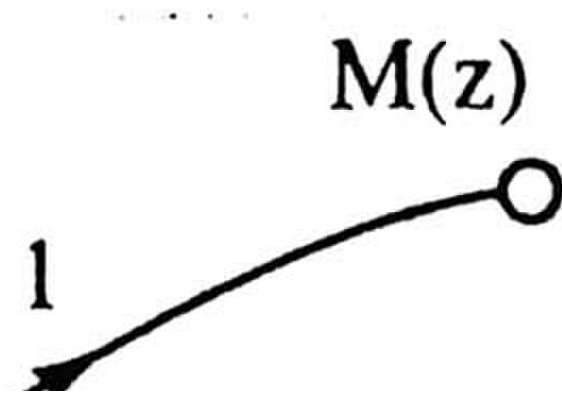
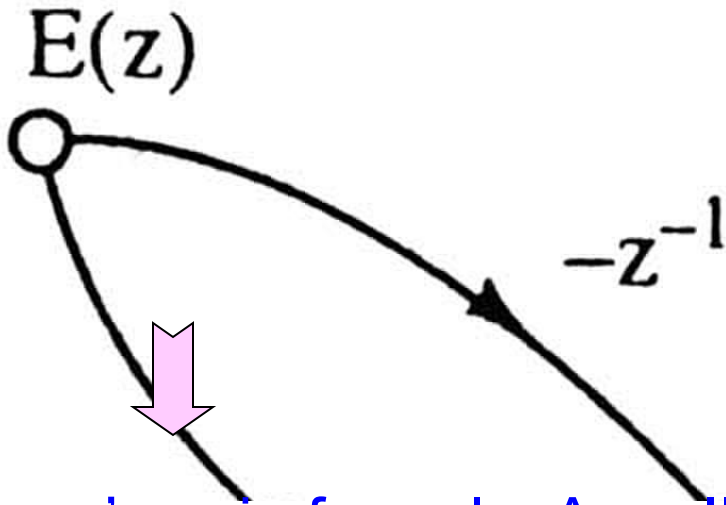
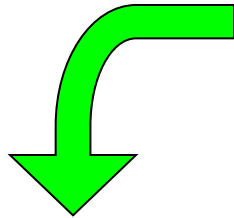


Figure 2-3 Simulation diagram for the difference equation used in Example 2.8.

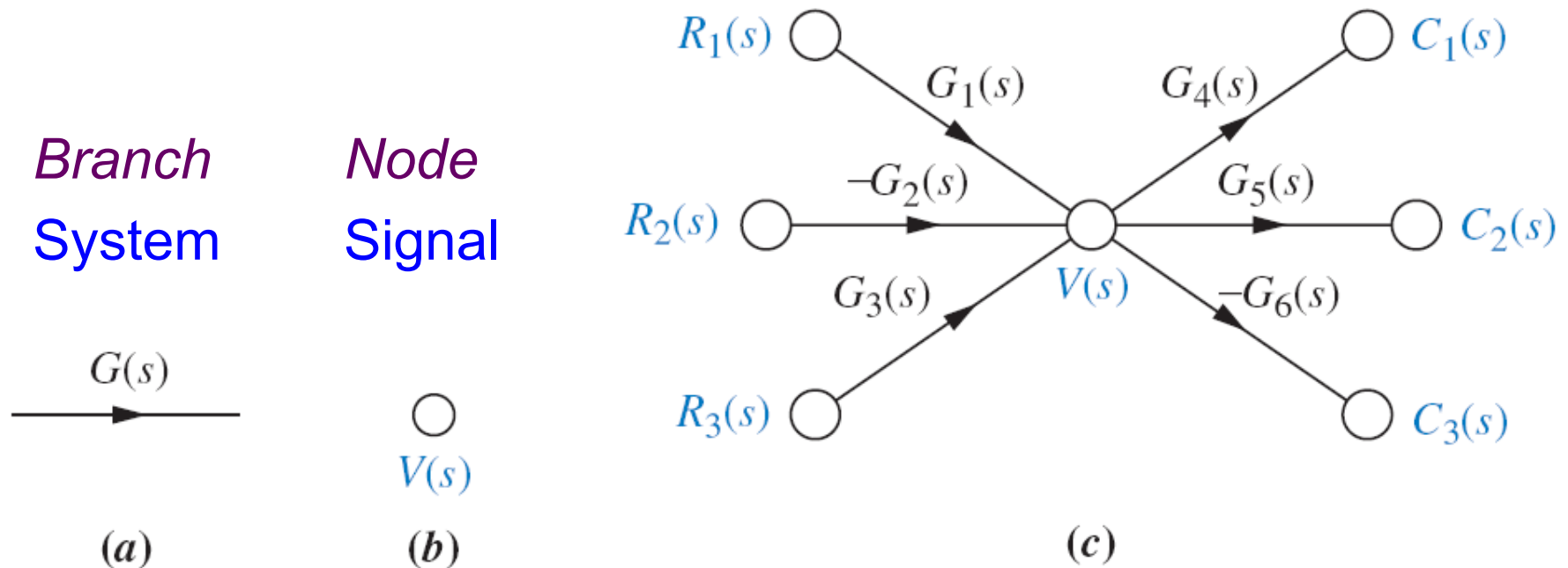


Mason's gain formula: App. II (page 626)

SIGNAL-FLOW GRAPHS

FIGURE 5.17 Signal-flow graph components: **a.** system; **b.** signal; **c.** interconnection of systems and signals

Each signal is the sum of signals **flowing into it**.



MASON'S RULE

- DEFINITIONS

- Loop gain

1. $G_2(s)H_1(s)$ (5.25a)

2. $G_4(s)H_2(s)$ (5.25b)

3. $G_4(s)G_5(s)H_3(s)$ (5.25c)

4. $G_4(s)G_6(s)H_3(s)$ (5.25d)

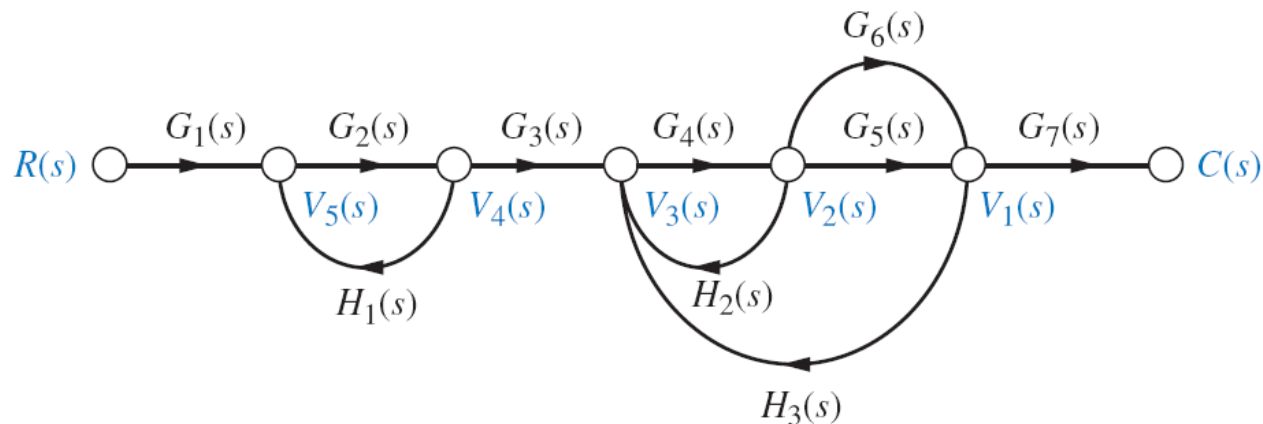


FIGURE 5.20 Signal-flow graph for demonstrating Mason's rule

- Forward-path gain

1. $G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)G_7(s)$ (5.26a)

2. $G_1(s)G_2(s)G_3(s)G_4(s)G_6(s)G_7(s)$ (5.26b)

- Nontouching loops: $G_2(s)H_1(s)$ does not touch other three loops.

- Nontouching-loop gain (*taken two at a time*)

1. $[G_2(s)H_1(s)][G_4(s)H_2(s)]$ (5.27a)

2. $[G_2(s)H_1(s)][G_4(s)G_5(s)H_3(s)]$ (5.27b)

3. $[G_2(s)H_1(s)][G_4(s)G_6(s)H_3(s)]$ (5.27c)



- MASON'S RULE

$$G(s) = \frac{C(s)}{R(s)} = \frac{\sum_k T_k \Delta_k}{\Delta}$$



(5.28)

EXAMPLE 5.7

Transfer function via Mason's rule

Problem: Find the transfer function, $C(s)/R(s)$, for the signal-flow graph in Figure 5.21.

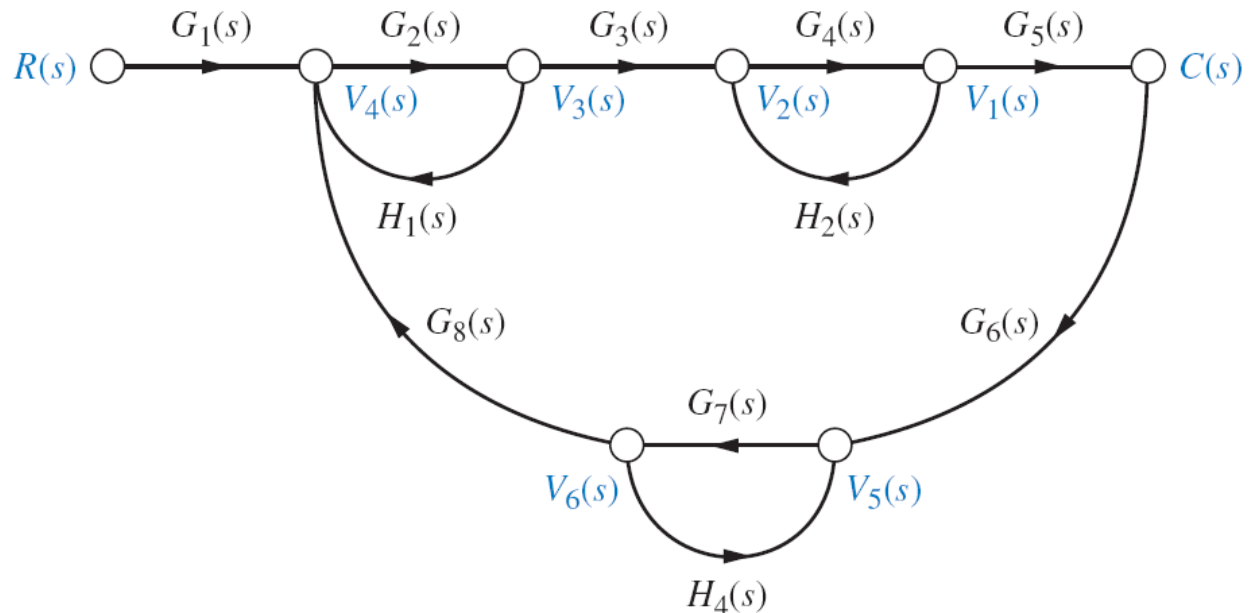


FIGURE 5.21 Signal-flow graph for Example 5.7

EXAMPLE 5.7

SOLUTION: First, identify the forward-path gains. In this example there is only one:

$$G_1(s)G_2(s)G_3(s)G_4(s)G_5(s) \quad \Rightarrow \quad T_1 \quad (5.29)$$

Second, identify the loop gains. There are four, as follows:

$$1. G_2(s)H_1(s) \quad (5.30a)$$

$$2. G_4(s)H_2(s) \quad (5.30b)$$

$$3. G_7(s)H_4(s) \quad (5.30c)$$

$$4. G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s) \quad (5.30d)$$

Third, identify the nontouching loops taken two at a time. From Eqs. (5.30) and Figure 5.21, we can see that loop 1 does not touch loop 2, loop 1 does not touch loop 3, and loop 2 does not touch loop 3. Notice that loops 1, 2, and 3 all touch loop 4. Thus, the combinations of nontouching loops taken two at a time are as follows:

$$\text{Loop 1 and loop 2: } G_2(s)H_1(s)G_4(s)H_2(s) \quad (5.31a)$$

$$\text{Loop 1 and loop 3: } G_2(s)H_1(s)G_7(s)H_4(s) \quad (5.31b)$$

$$\text{Loop 2 and loop 3: } G_4(s)H_2(s)G_7(s)H_4(s) \quad (5.31c)$$

EXAMPLE 5.7

Finally, the *nontouching loops taken three at a time* are as follows:

$$\text{Loops 1, 2, and 3: } G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s) \quad (5.32)$$

Now, from Eq. (5.28) and its definitions, we form Δ and Δ_k . Hence,

$$\begin{aligned} \Delta = 1 &- [G_2(s)H_1(s) + G_4(s)H_2(s) + G_7(s)H_4(s) \\ &+ G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)] \\ &+ [G_2(s)H_1(s)G_4(s)H_2(s) + G_2(s)H_1(s)G_7(s)H_4(s) \\ &+ G_4(s)H_2(s)G_7(s)H_4(s)] \\ &- [G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)] \end{aligned} \quad (5.33)$$

EXAMPLE 5.7

We form Δ_k by eliminating from Δ the loop gains that touch the k th forward path:

$$\Delta_1 = 1 - G_7(s)H_4(s) \quad (5.34)$$

Expressions (5.29), (5.33), and (5.34) are now substituted into Eq. (5.28), yielding the transfer function:

$$\underline{G(s) = \frac{T_1 \Delta_1}{\Delta} = \frac{[G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)][1 - G_7(s)H_4(s)]}{\Delta}} \quad (5.35)$$

Since there is only one forward path, $G(s)$ consists of only one term, rather than a sum of terms, each coming from a forward path.

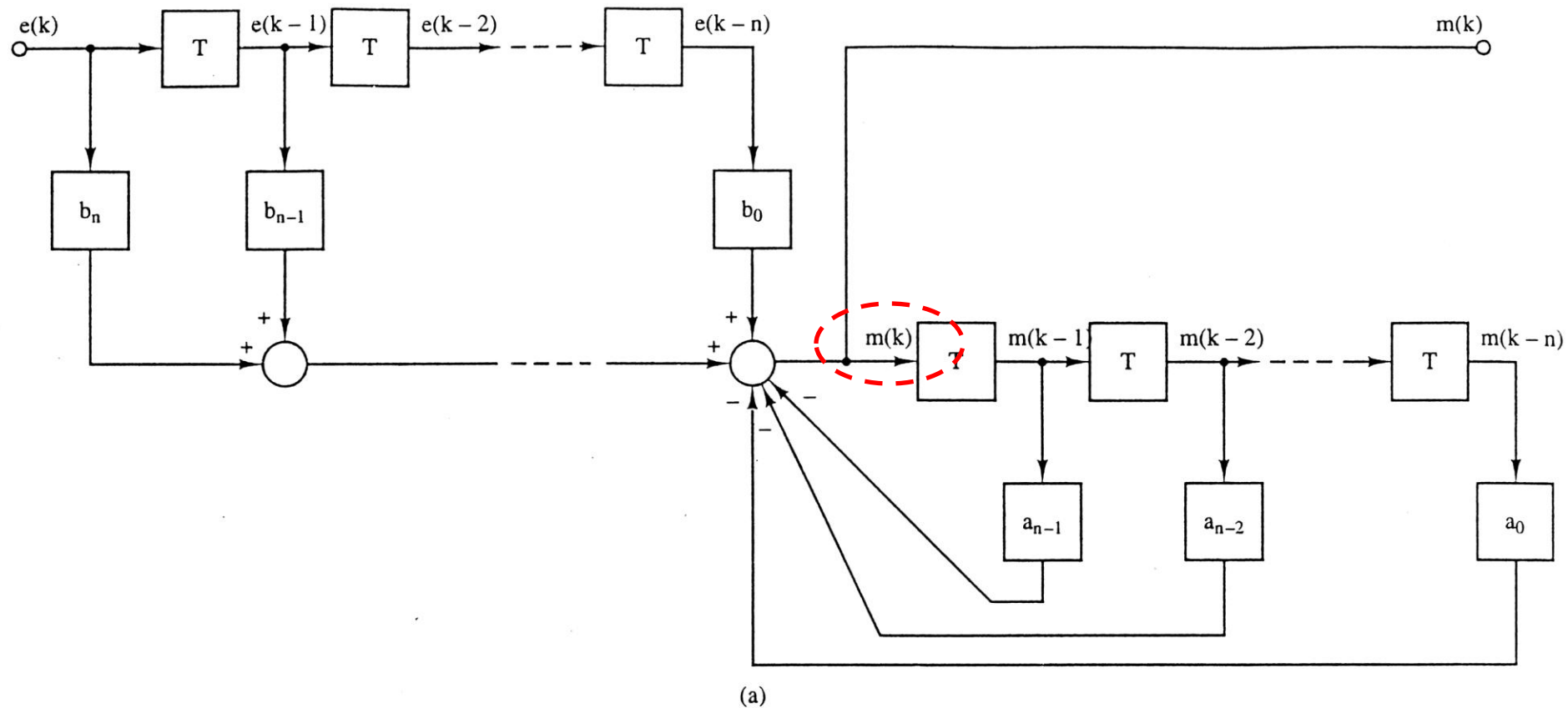
More: Check the “feedback control form”.

nth-order DE

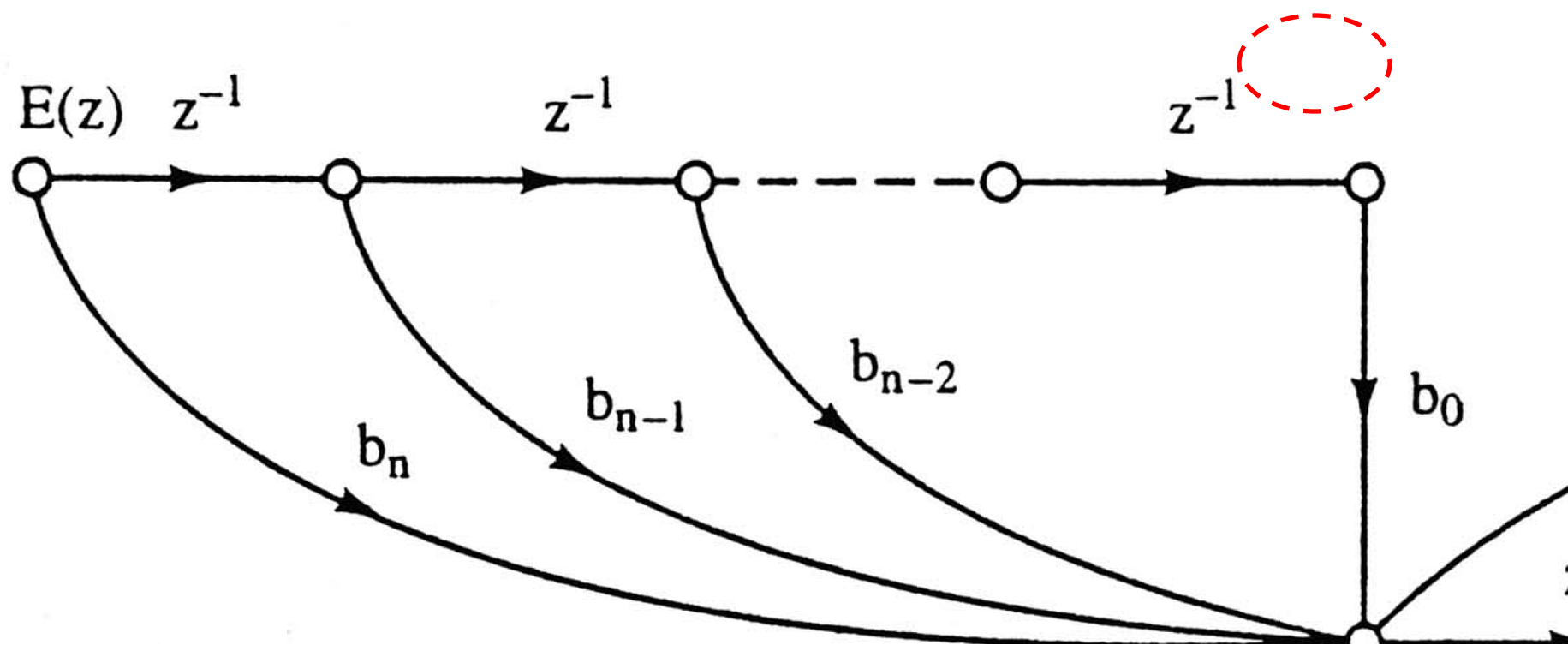


$$\frac{M(z)}{E(z)} = \frac{b_n + b_{n-1}z^{-1} + \dots + b_0z^{-n}}{1 + a_{n-1}z^{-1} + \dots + a_0z^{-n}} \quad (2.43)$$

(2.43)



Simulation diagram (non-minimal representation: $2n$ delays)



2.8 State Variables

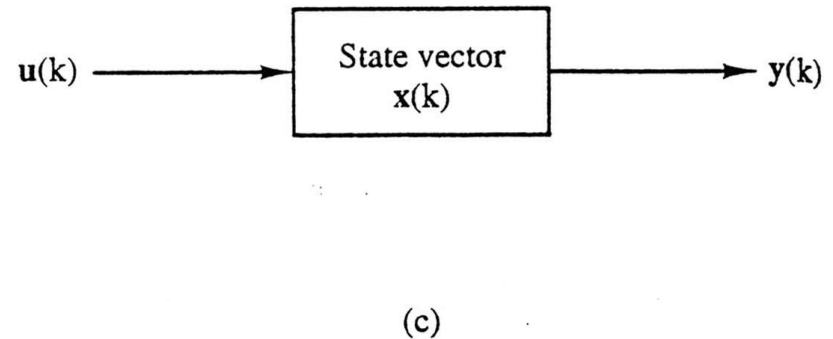
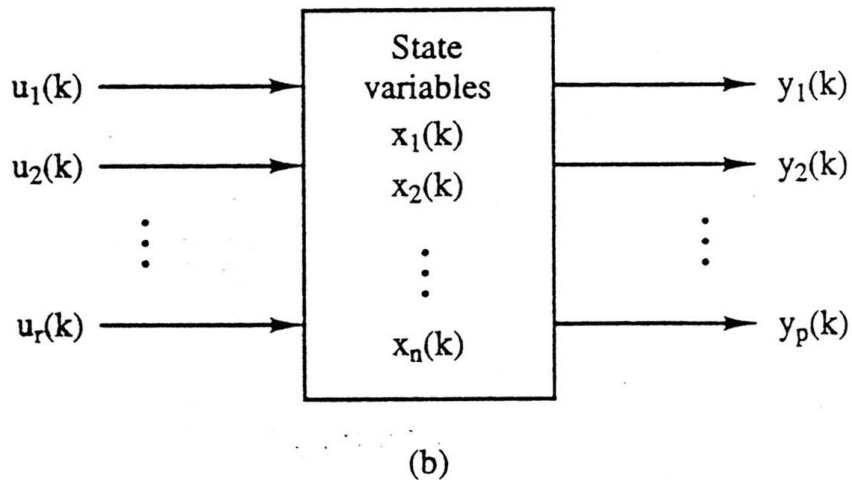
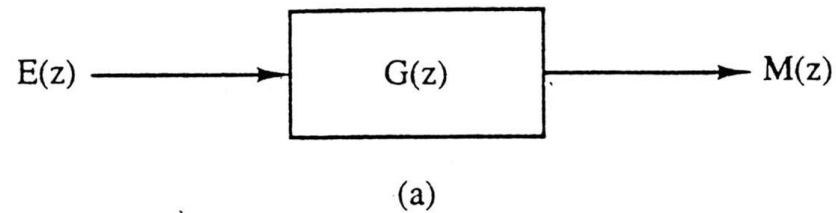


Figure 2-8 Representations of system dynamics: (a) z-transfer function representation; (b) state-variable representation; (c) state-vector representation.

$$\begin{cases} x(k+1) = f[x(k), u(k)] \\ y(k) = g[x(k), u(k)] \end{cases} \quad (2.45-46)$$

If the system is linear

$$\begin{cases} x(k+1) = A(k)x(k) + B(k)u(k) \\ y(k) = C(k)x(k) + D(k)u(k) \end{cases} \quad (2.47-48)$$

If the system is time invariant

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{cases}$$

D : Feedforward matrix

(2.49-50)

It is desired to find a state-variable model of the system described by the difference equation

補充: SFG (p.55)

$$x_1(k) = y(k)$$
$$x_2(k) = x_1(k + 1) = y(k + 1)$$
[illegible]

or, from these equations, we write

$$x_1(k+1) = x_2(k)$$

$$x_2(k+1) = -0.72x_1(k) + 1.7x_2(k) + u(k)$$

$$y(k) = x_1(k)$$

We may express these equations in vector-matrix form of (2-49) and (2-50) as

$$\mathbf{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.72 & 1.7 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$
$$y(k) = [1 \quad 0] \mathbf{x}(k)$$

Phase variable canonical form / **control** canonical form

Given the transfer function

$$G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0} \quad (2.51)$$

Write as

$$\frac{Y(z)}{U(z)} = G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0} \frac{E(z)}{E(z)} \quad (2.52)$$

let

$$Y(z) = (b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0)E(z)$$

sin(ωt) vs. cos(ωt) 相位

$$U(z) = (z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0)E(z) \quad (2.53-54)$$

Phase variables

Define the state variables

$$\begin{aligned} x_1(k) &= e(k) \\ x_2(k) &= x_1(k+1) = e(k+1) \\ x_3(k) &= x_2(k+1) = e(k+2) \\ &\vdots \\ x_n(k) &= x_{n-1}(k+1) = e(k+n-1) \end{aligned}$$

(2.55)

Obtain the state equation

$$\begin{aligned} x_1(k+1) &= x_2(k) \\ x_2(k+1) &= x_3(k) \\ x_3(k+1) &= x_4(k) \\ &\vdots \\ x_n(k+1) &= -a_0x_1(k) - a_1x_2(k) - a_2x_3(k) \cdots - a_{n-1}x_n(k) + u(k) \end{aligned}$$

(2.56)

58/90

In matrix form is

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ \vdots \\ x_n(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & \cdots & 0 \\ & & & \vdots & & & \\ -a_0 & -a_1 & -a_2 & -a_3 & -a_4 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix}$$

About b_n in (2.43),
see Slide 66.

(2.57, 59)

State Equations

or simply

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) \end{cases}$$

(2.58, 60)

Signal Flow Graph Representation

The form can be derived as

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_{n-1}z^{-1} + b_{n-2}z^{-2} + \cdots + b_1z^{1-n} + b_0z^{-n} E(z)}{1 + a_{n-1}z^{-1} + \cdots + a_1z^{1-n} + a_0z^{-n} E(z)} \quad (2.61)$$

$$\begin{cases} Y(z) = (b_{n-1}z^{-1} + b_{n-2}z^{-2} + \cdots + b_1z^{1-n} + b_0z^{-n})E(z) \\ U(z) = (1 + a_{n-1}z^{-1} + \cdots + a_1z^{1-n} + a_0z^{-n})E(z) \end{cases} \quad (2.62-63)$$

Can be written as $E(z) = U(z) - a_{n-1}z^{-1}E(z) - \cdots - a_1z^{1-n}E(z) - a_0z^{-n}E(z)$ (2.64)

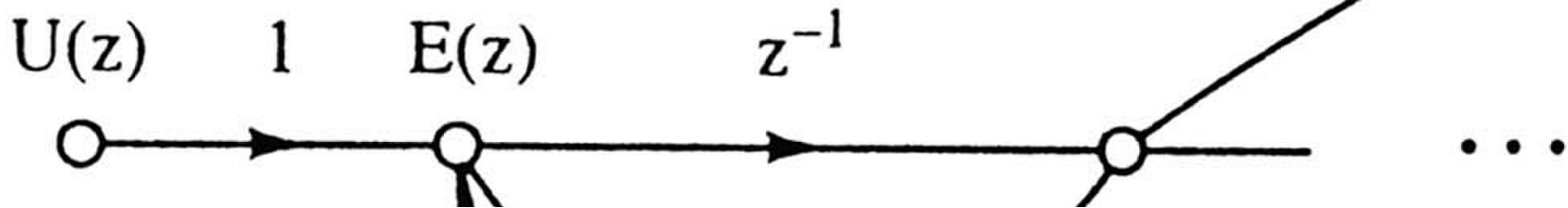
Signal Flow Graph Representation

$$G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0}$$

(2.51)

Y (z): see (2.62)

E (z): see (2.64)



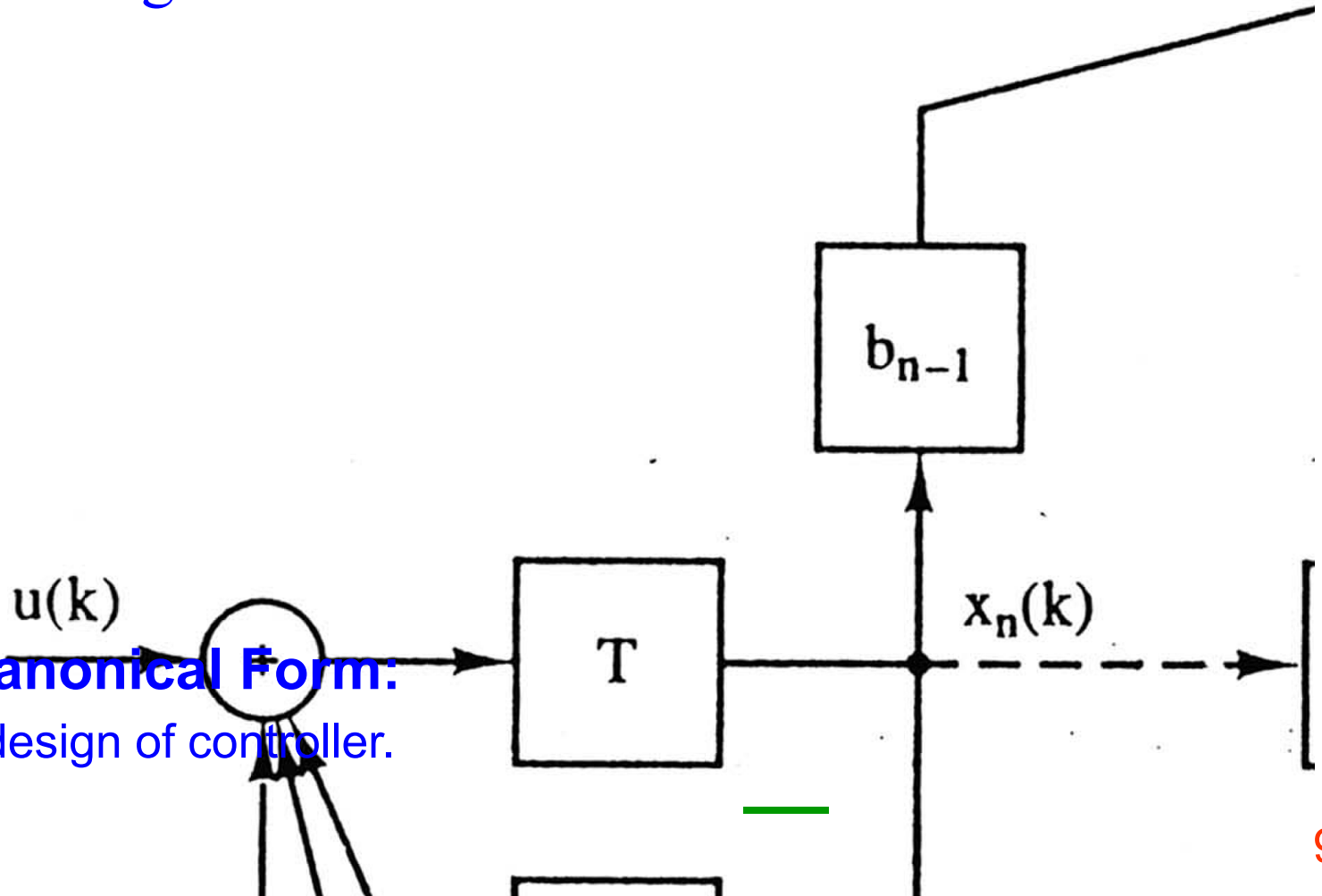
$$G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0}$$

(2.51)

(a)

Simulation Diagram

Control Canonical Form:
Use in the design of controller.



Observer Canonical Form:

Use in the design of observer.

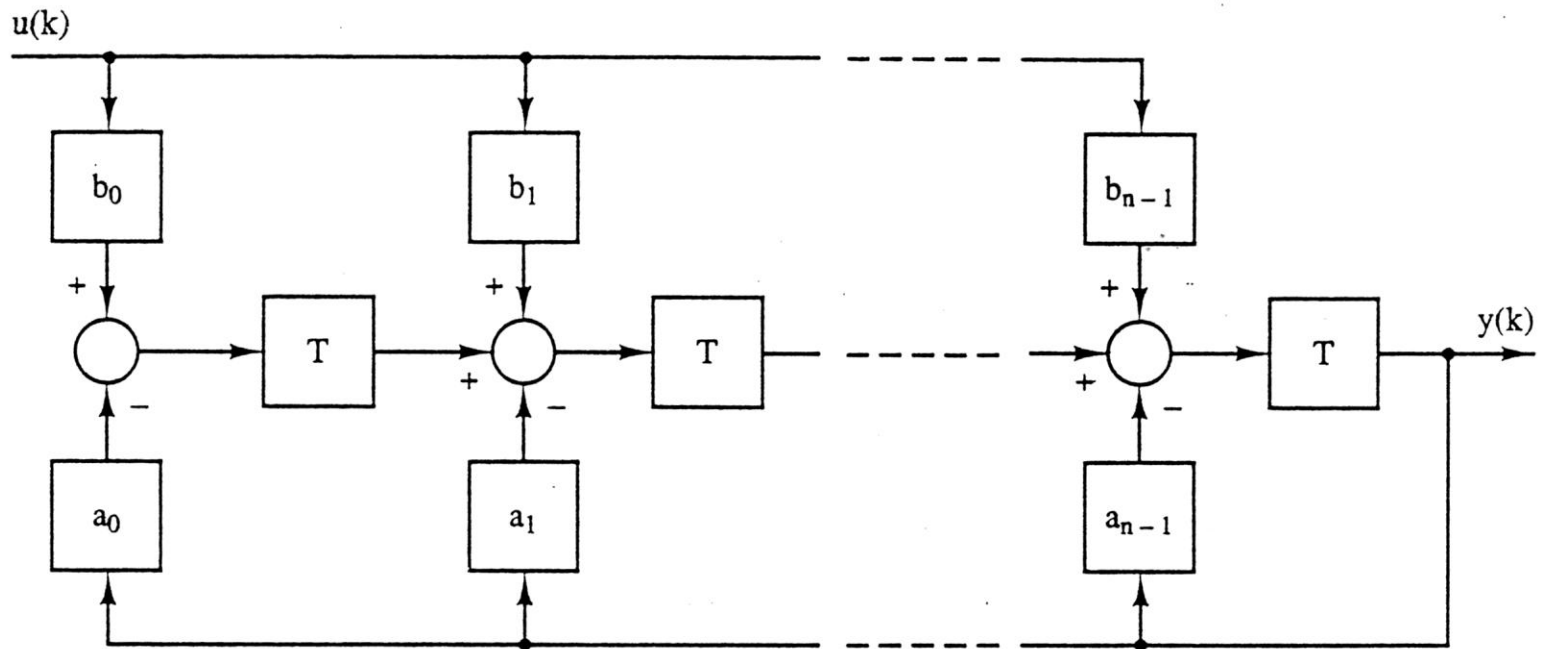


Figure 2-10 Observer canonical form.

$$G(z) = \frac{b_{n-1}z^{n-1} + b_{n-2}z^{n-2} + \cdots + b_1z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0} \quad (2.51)$$

Example 2.17

Given the following function, we want to derive a signal flow graph and corresponding state equations:

$$G(z) = \frac{Y(z)}{U(z)} = \frac{z^2 + 2z + 1}{z^3 + 2z^2 + z + \frac{1}{2}}$$

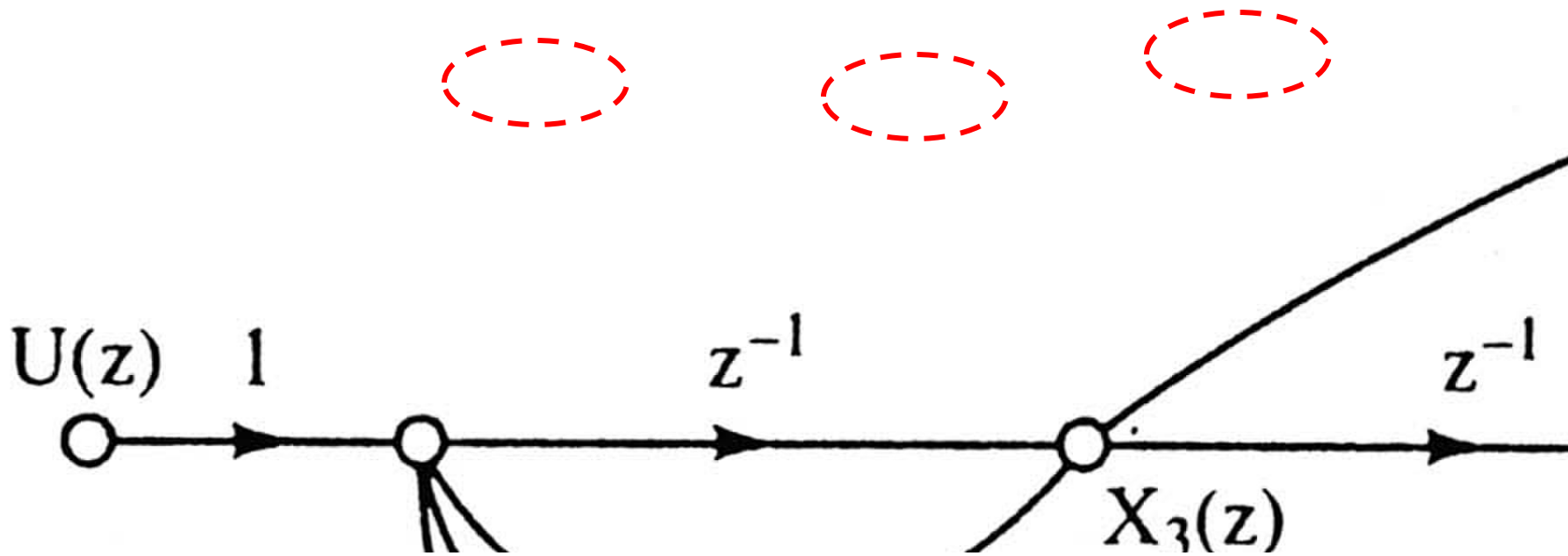
Comparing this expression with (2-51) and Figure 2-9a indicates that the signal flow graph can be derived by inspection, as shown in Figure 2-11. Note that the states are then chosen to be the delay outputs. The state equations can now be derived from Figure 2-11 or from a comparison of the given transfer function with equations (2-51), (2-57), and (2-59). The resulting state equations are

Phase variables

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{2} & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [1 \quad 2 \quad 1] \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

Control Canonical Form:
Use in the design of controller.



The same order (num. & denom.)

Example 2.18

Given the following transfer function, we wish to derive the state equations.

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_2 z^2 + b_1 z + b_0}{z^2 + a_1 z + a_0}$$

In this case, the order of the numerator is equal to that of the denominator. First we express the transfer function as

$$\frac{Y(z)}{U(z)} = \frac{b_2 + b_1 z^{-1} + b_0 z^{-2}}{1 + a_1 z^{-1} + a_0 z^{-2}} \frac{E(z)}{E(z)}$$

b_n is not zero!

$$\frac{M(z)}{E(z)} = \frac{b_n + b_{n-1} z^{-1} + \dots + b_0 z^{-n}}{1 + a_{n-1} z^{-1} + \dots + a_0 z^{-n}} \quad (2.43)$$

We see from Mason's gain formula that the signal flow graph for this transfer function is as shown in Figure 2-12. The states are shown on the flow graph, and the equations are seen to be

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

At this point it is important to note that the flow graph in Figure 2-12 is slightly different from the form presented earlier; that is, there is a direct path from $E(z)$ to $Y(z)$. From the flow graph the equation for the output is

$$Y(z) = b_0 X_1(z) + b_1 X_2(z) + b_2 E(z)$$

Feedforward control

But

$$E(z) = U(z) - a_0 X_1(z) - a_1 X_2(z)$$

From these equations we obtain

$$Y(z) = b_2 U(z) + (b_0 - b_2 a_0) X_1(z) + (b_1 - b_2 a_1) X_2(z)$$

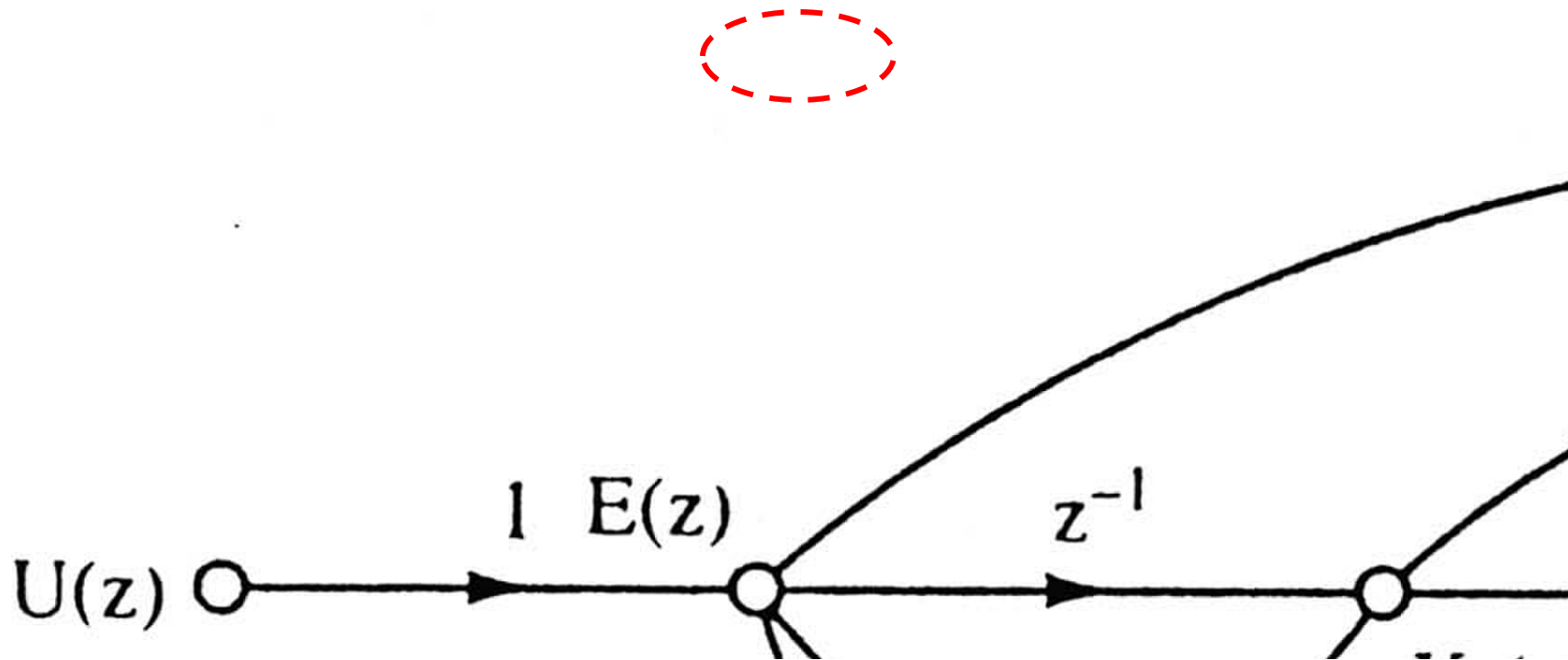
Hence the output equation is

$$y(k) = [(b_0 - b_2 a_0) \quad (b_1 - b_2 a_1)] \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + b_2 u(k)$$

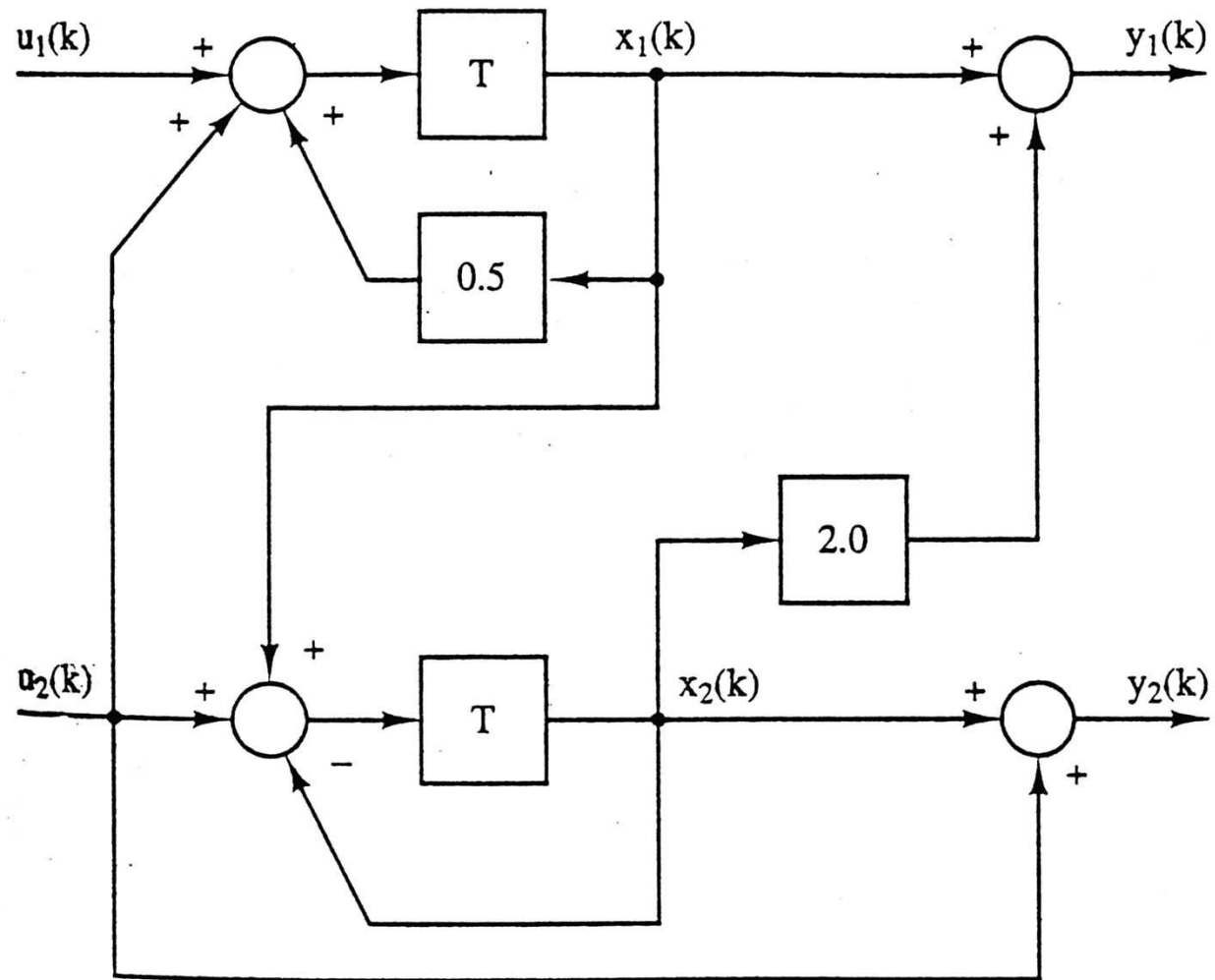
$D = b_2$

Note the difference in this equation and (2-59).

D: Feedforward matrix



Example 2.19
2 inputs



2 Approaches:

- 1) TF $\rightarrow$ SS: use Eq. (2-52)~(2-59)
- 2) TF $\rightarrow$ diagrams $\rightarrow$ SS

Figure 2-13 Simulation diagram for a multivariable system.

Approach 2

Procedure to Obtain State Models

- 1) Draw a block diagram or signal flow chart.
- 2) Assign a state variable to each delay output, $x(k)$.
- 3) Write the equation for each delay input, $x(k+1)$, and each system output, $y(k)$, in terms of only the delay outputs, $x(k)$, and the system input, $u(k)$.

2.9 Other State-Variable Formulations

Example 2.20

Consider the difference equation of Example 2.16:

$$y(k + 2) = u(k) + 1.7y(k + 1) - 0.72y(k)$$

One state-variable model was derived in that example. Two additional models are derived here. Taking the z -transform of the difference equation yields

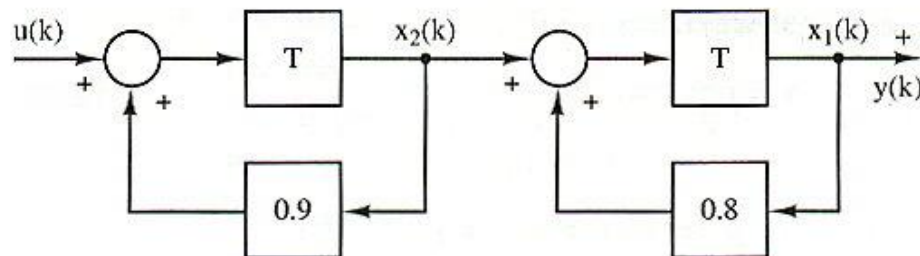
$$\frac{Y(z)}{U(z)} = \frac{1}{z^2 - 1.7z + 0.72}$$

This transfer function can be expressed as

$$\frac{Y(z)}{U(z)} = \left[\frac{1}{z - 0.9} \right] \left[\frac{1}{z - 0.8} \right]$$

A simulation diagram for this transfer function is given in Figure 2-14a. From this figure we write the state equations

$$\mathbf{x}(k + 1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$
$$y(k) = [1 \quad 0] \mathbf{x}(k)$$



(a)

Cascade

Also, the system transfer function can be expressed, through partial-fraction expansion, as

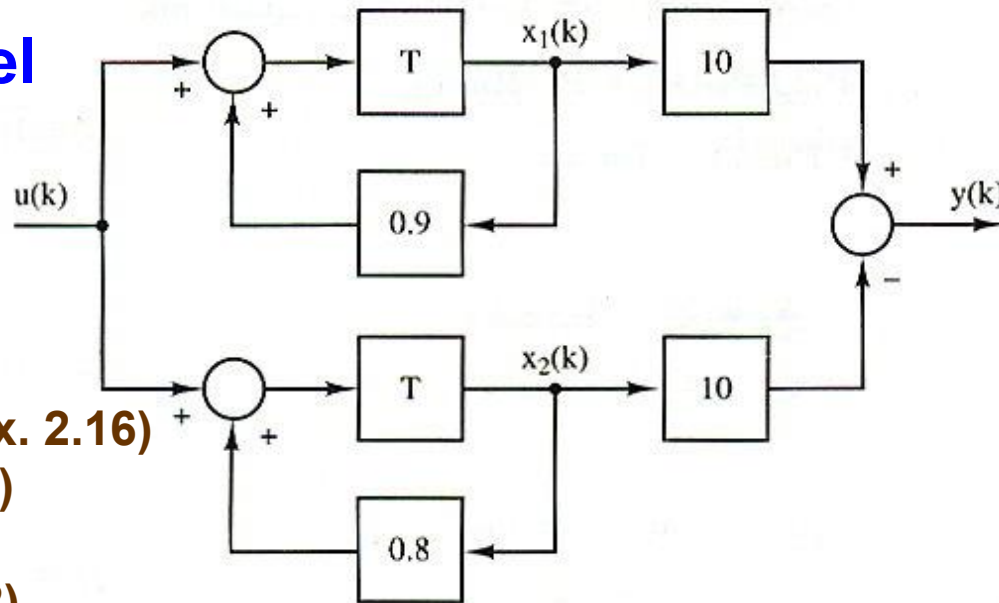
$$\frac{Y(z)}{U(z)} = \frac{10}{z - 0.9} + \frac{-10}{z - 0.8}$$

A simulation diagram for this transfer function is given in Figure 2-14b. From this figure we write the state equations

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.8 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [10 \quad -10] \mathbf{x}(k)$$

Parallel



Different SS models

- 1) Phase variable (Ex. 2.16)
- 2) Cascade (Ex. 2.20)
- 3) Parallel (Ex. 2.20)
- 4) Diagonal (Ex. 2.22)
- 5) Any More... (Ex. 2.21)

Similarity Transformations

Consider the state equation

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{cases} \quad (2.67)$$

Apply the linear transformation $x(k) = Pw(k)$ (2.68)

yield

$$\begin{cases} w(k+1) = P^{-1}APw(k) + P^{-1}Bu(k) \\ y(k) = CPw(k) + Du(k) \end{cases} \quad (2.69)$$

These equations can be expressed

$$\begin{cases} w(k+1) = A_w w(k) + B_w u(k) \\ y(k) = C_w w(k) + D_w u(k) \end{cases} \quad (2.70)$$

Where

$$\begin{cases} A_w = P^{-1}AP, & B_w = P^{-1}B \\ C_w = CP, & D_w = D \end{cases} \quad (2.71)$$

The characteristic equation is defined by $|zI - A| = 0$ (2.72)

$$|zI - A| = (z - z_1)(z - z_2) \cdots (z - z_n) = 0 \quad (2.73)$$

and
$$|zI - A_w| = |zI - P^{-1}AP| = |zP^{-1}IP - P^{-1}AP| = |P^{-1}||zI - A||P| = |zI - A| \quad (2.74)$$

Similarity transformation has the following properties

1. As shown above, the characteristic values of the matrix are unchanged under the transformation.
2. The determinant of A_w (which is the product of characteristic values) is equal to the determinant of A :

$$|A_w| = |P^{-1}AP| = |P^{-1}||A||P| = |A| = z_1 z_2 \cdots z_n$$

3. The trace of A_w is equal to the trace of A :

Trace (跡): see P639. $\text{tr } A_w = \text{tr } A = z_1 + z_2 + \cdots + z_n$

since the trace of a matrix, which is the sum of the diagonal elements, is equal to the sum of its characteristic values. The latter property follows immediately from the first property.

4. As is shown in Section 2.10,

Unique TF
$$\underline{C[zI - A]^{-1}B + D = C_w[zI - A_w]^{-1}B_w + D_w}$$

Example 2.21

For the system of Example 2.20, one state-variable model given is

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$
$$y(k) = [1 \quad 0] \mathbf{x}(k)$$

We arbitrarily choose a linear-transformation matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

The inverse of $\mathbf{P}$ is given by

$$\mathbf{P}^{-1} = \frac{[\text{Cof}[\mathbf{P}]]^T}{|\mathbf{P}|}$$

where $[\cdot]^T$ indicates the transpose, and $\text{Cof}[\mathbf{P}]$ denotes the matrix of cofactors of $\mathbf{P}$.
Thus

$$\text{Cof}[\mathbf{P}] = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \text{Cofactor (餘因子): See P640.}$$

and

$$|\mathbf{P}| = 2$$

Then

$$\mathbf{P}^{-1} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Thus, from (2-71),

$$\begin{aligned}\mathbf{A}_w &= \mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 1.8 & 0.2 \\ 0.9 & 0.9 \end{bmatrix} = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix} \\ \mathbf{B}_w &= \mathbf{P}^{-1} \mathbf{B} = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \\ \mathbf{C}_w &= \mathbf{C} \mathbf{P} = [1 \quad 0] \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = [1 \quad -1]\end{aligned}$$

Then the new state equations are

$$\begin{aligned}\mathbf{w}(k+1) &= \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix} \mathbf{w}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} u(k) \\ y(k) &= [1 \quad -1] \mathbf{w}(k)\end{aligned}$$

Note that

$$|z\mathbf{I} - \mathbf{A}| = \begin{vmatrix} z - 0.8 & -1 \\ 0 & z - 0.9 \end{vmatrix} = z^2 - 1.7z + 0.72 = (z - 0.8)(z - 0.9)$$

and

$$|z\mathbf{I} - \mathbf{A}_w| = \begin{vmatrix} z - 1.35 & -0.55 \\ 0.45 & z - 0.35 \end{vmatrix} = z^2 - 1.7z + 0.72$$

Hence the characteristic values are $z_1 = 0.8$ and $z_2 = 0.9$. Also,

$$\underline{|\mathbf{A}| = |\mathbf{A}_w| = 0.72 = z_1 z_2}$$

and

$$\underline{\text{tr } \mathbf{A} = \text{tr } \mathbf{A}_w = 1.7 = z_1 + z_2}$$

Modal Matrix 形式矩陣

If a system has distinct characteristic values, we may derive a state-variable model in which the system matrix is diagonal. Consider a vector $\mathbf{m}_i$ and a scalar z_i defined by the equation

$$\mathbf{A}\mathbf{m}_i = z_i \mathbf{m}_i \quad (2-76)$$

where $\mathbf{m}_i = [m_{1i} \ m_{2i} \ \cdots \ m_{ni}]^T$. We express (2-76) as

$$(z_i \mathbf{I} - \mathbf{A})\mathbf{m}_i = \mathbf{0}$$

For a nontrivial solution for this equation to exist, it is required that

$$|z_i \mathbf{I} - \mathbf{A}| = 0$$

Hence, from (2-72), it is seen that z_i is a characteristic value (also called an eigenvalue) of $\mathbf{A}$. The vector $\mathbf{m}_i$ is called a characteristic vector, or eigenvector, of $\mathbf{A}$. From (2-76) we can construct the equation

$$\mathbf{A}[\mathbf{m}_1 \ \mathbf{m}_2 \ \cdots \ \mathbf{m}_n] = [\mathbf{m}_1 \ \mathbf{m}_2 \ \cdots \ \mathbf{m}_n] \begin{bmatrix} z_1 & 0 & \cdot & 0 \\ 0 & z_2 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & z_n \end{bmatrix}$$

or

$$\mathbf{A}\mathbf{M} = \mathbf{M}\mathbf{\Lambda}$$

where $\mathbf{M}$, called the *modal matrix*, is composed of the characteristic vectors as columns, and $\mathbf{\Lambda}$ is a diagonal matrix with the characteristic values of $\mathbf{A}$ as the diagonal elements. The characteristic vectors are linearly independent provided the characteristic values are distinct. Hence we can write

$$\mathbf{\Lambda} = \mathbf{M}^{-1} \mathbf{A} \mathbf{M} \quad (2-77)$$

Then, in the general similarity transformation, $\mathbf{P}$ is equal to $\mathbf{M}$, the modal matrix. An example will be given to illustrate this procedure.

Example 2.22

For the system of Example 2.21

$$\mathbf{x}(k+1) = \begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [1 \quad 0] \mathbf{x}(k)$$

From Example 2.21, the characteristic values are $z_1 = 0.8$ and $z_2 = 0.9$. From (2-76),

$$\begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix} = 0.8 \begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix}$$

Thus

$$0.8m_{11} + m_{21} = 0.8m_{11}$$

$$0.9m_{21} = 0.8m_{21}$$

Hence $m_{21} = 0$ and m_{11} is arbitrary. Let $m_{11} = 1$. Also,

$$\begin{bmatrix} 0.8 & 1 \\ 0 & 0.9 \end{bmatrix} \begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix} = 0.9 \begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix}$$

Then m_{22} is seen to be arbitrary; let $m_{22} = 1$. Then $m_{12} = 10$. The modal matrix and its inverse are

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$$

and

$$\mathbf{M}^{-1} = \begin{bmatrix} 1 & -10 \\ 0 & 1 \end{bmatrix}$$

From (2-71) and (2-77),

$$\mathbf{A} = \mathbf{M}^{-1} \mathbf{A} \mathbf{M} = \begin{bmatrix} 0.8 & 0 \\ 0 & 0.9 \end{bmatrix}$$

$$\mathbf{B}_w = \mathbf{M}^{-1} \mathbf{B} = \begin{bmatrix} -10 \\ 1 \end{bmatrix}$$

$$\mathbf{C}_w = \mathbf{C} \mathbf{M} = [1 \quad 10]$$

Hence the new state model is given by

$$\begin{aligned} \mathbf{w}(k+1) &= \begin{bmatrix} 0.8 & 0 \\ 0 & 0.9 \end{bmatrix} \mathbf{w}(k) + \begin{bmatrix} -10 \\ 1 \end{bmatrix} u(k) \\ y(k) &= [1 \quad 10] \mathbf{w}(k) \end{aligned}$$

2.10 Transfer Functions

- Two Approaches to obtain TF via SS
 - 1) SS $\rightarrow$ Diagrams $\rightarrow$ TF (by Mason's rule)
 - 2) SS $\rightarrow$ TF (by z-Transform)

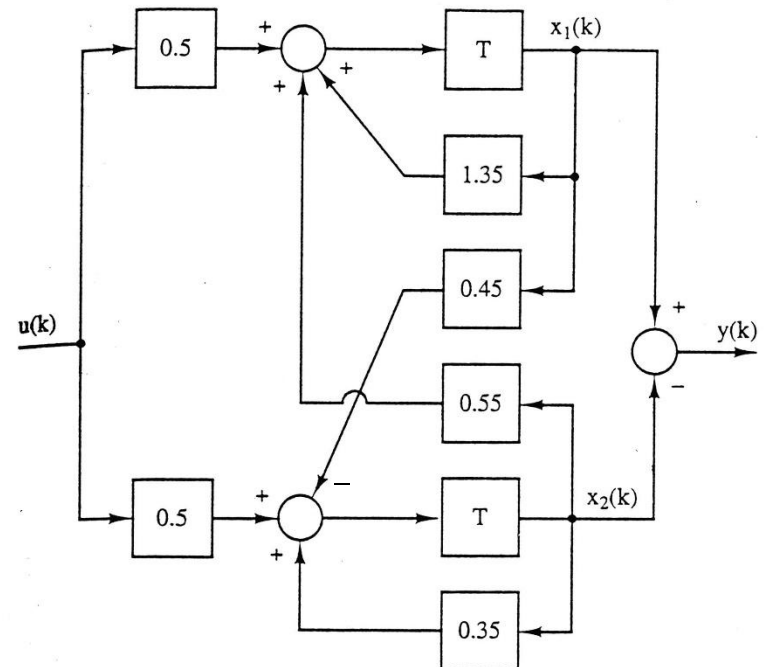
Example 2.23

Approach 1

Consider the state model derived in Example 2.21.

$$\mathbf{x}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} u(k)$$
$$y(k) = [1 \quad -1] \mathbf{x}(k)$$

A simulation diagram, constructed from these equations, is shown in Figure 2-15.



Application of Mason's gain formula to this figure yields

$$\frac{Y(z)}{U(z)} = \frac{0.5z^{-1}(1 - 0.35z^{-1}) - 0.5z^{-1}(1 - 1.35z^{-1}) + (0.5)(0.45)z^{-2} + (0.5)(0.55)z^{-2}}{1 - 1.35z^{-1} - 0.35z^{-1} + (0.45)(0.55)z^{-2} + (1.35z^{-1})(0.35z^{-1})}$$
$$= \frac{z^{-2}}{1 - 1.7z^{-1} + 0.72z^{-2}} = \frac{1}{z^2 - 1.7z + 0.72}$$

SS $\rightarrow$ TF (by z-Transform) Eliminate $X(z)$

Another approach for obtaining the system transfer function from the state equations

$$x(k+1) = Ax(k) + Bu(k) \quad (2.78)$$

Taking the z-transform yields $zX(z) - \underline{zx(0)} = AX(z) + BU(z)$ (2.79)

Ref P37.  Ignore initial conditions

$$[zI - A]X(z) = BU(z) \quad (2.80)$$

$$X(z) = [zI - A]^{-1} BU(z) \quad (2.81)$$

also $y(k) = Cx(k) + Du(k)$ (2.82)

Taking the z-transform yields $Y(z) = CX(z) + DU(z)$ (2.83)

$$Y(z) = [C[zI - A]^{-1}B + D]U(z)$$

The system transfer function is



$$G(z) = C[zI - A]^{-1}B + D$$

(2.84)

Example 2.24

Consider again the state equations of Example 2.23.

Approach 2

$$\mathbf{x}(k+1) = \begin{bmatrix} 1.35 & 0.55 \\ -0.45 & 0.35 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} u(k)$$
$$y(k) = [1 \quad -1] \mathbf{x}(k)$$

Now

$$[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z - 1.35 & -0.55 \\ 0.45 & z - 0.35 \end{bmatrix}$$

Thus

$$|z\mathbf{I} - \mathbf{A}| = z^2 - 1.7z + 0.72$$

Also,

$$\text{Cof}[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z - 0.35 & -0.45 \\ 0.55 & z - 1.35 \end{bmatrix}$$

Then

$$[z\mathbf{I} - \mathbf{A}]^{-1} = \frac{[\text{Cof}[z\mathbf{I} - \mathbf{A}]]^T}{|z\mathbf{I} - \mathbf{A}|} = \frac{1}{z^2 - 1.7z + 0.72} \begin{bmatrix} z - 0.35 & 0.55 \\ -0.45 & z - 1.35 \end{bmatrix}$$

From (2-84), since $D = 0$,

$$\underline{G(z) = \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1} \mathbf{B}}$$

$$= \frac{1}{z^2 - 1.7z + 0.72} [1 \quad -1] \begin{bmatrix} z - 0.35 & 0.55 \\ -0.45 & z - 1.35 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

$$= \frac{1}{z^2 - 1.7z + 0.72} [1 \quad -1] \begin{bmatrix} 0.5z + 0.1 \\ 0.5z - 0.9 \end{bmatrix}$$

$$= \underline{\frac{1}{z^2 - 1.7z + 0.72}}$$

2.11 Solutions of The State Equations

1) Recursive solution

The system equation
$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) + Du(k) \end{cases} \quad (2.86-87)$$

that
$$x(1) = Ax(0) + Bu(0) \quad x(2) = Ax(1) + Bu(1)$$

hence
$$x(2) = A(Ax(0) + Bu(0)) + Bu(1) = A^2x(0) + ABu(0) + Bu(1)$$

similarly
$$x(3) = A^3x(0) + A^2Bu(0) + ABu(1) + Bu(2)$$

Then, the general solution is given by
$$x(k) = A^k x(0) + \sum_{j=0}^{k-1} A^{(k-1-j)} Bu(j) \quad (2.88)$$

If we define $\Phi(k) = A^k$ **State transition/fundamental matrix**

狀態轉移/基本矩陣

$$\begin{cases} x(k) = \Phi(k)x(0) + \sum_{j=0}^{k-1} \Phi(k-1-j)Bu(j) \\ y(k) = C\Phi(k)x(0) + \sum_{j=0}^{k-1} C\Phi(k-1-j)Bu(j) + Du(k) \end{cases} \quad (2.89-90)$$

Example 2.25

Consider the transfer function

$$G(z) = \frac{(z + 3)}{(z + 1)(z + 2)}$$

Using the technique of Section 2.8, we write

$$\mathbf{x}(k + 1) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [3 \quad 1] \mathbf{x}(k)$$

Assume that the system is initially at rest so that $\mathbf{x}(0) = \mathbf{0}$, and that the input is a unit step; that is,

$$u(k) = 1, \quad k = 0, 1, 2, \dots$$

The recursive solution is obtained as follows:

$$\mathbf{x}(1) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}(0) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$y(1) = [3 \quad 1] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

Then

$$\begin{aligned} \mathbf{x}(2) &= \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x}(1) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(1) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (1) \\ &= \begin{bmatrix} 1 \\ -2 \end{bmatrix} \end{aligned}$$

$$y(2) = [3 \quad 1] \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 3 - 2 = 1$$

In a similar manner it can be shown that

$$\mathbf{x}(3) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, \quad y(3) = -1$$

and

$$\mathbf{x}(4) = \begin{bmatrix} 5 \\ -10 \end{bmatrix}, \quad y(4) = 5, \text{ etc.}$$

2) z-transform Method

The general solution to the state equation $x(k+1) = Ax(k) + Bu(k)$ (2.91)

Let $u(k) = 0$ Then the z-transform of this equation yields

$$zX(z) - zx(0) = AX(z) \quad (2.93)$$

$$X(z) = z[zI - A]^{-1} x(0) \quad (2.94)$$

Then take z-inverse $x(k) = z^{-1}[X(z)] = z^{-1}[z[zI - A]^{-1}]x(0)$ (2.95)

Compared with (2.89) $x(k) = \Phi(k)x(0) + \sum_{j=0}^{k-1} \Phi(k-1-j)Bu(j)$

$$\Phi(k) = z^{-1} \left[z[zI - A]^{-1} \right] \quad (2.96)$$

Recall: $\Phi(k) = A^k$

Example 2.26

For the state equations of Example 2.25,

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Then

$$[z\mathbf{I} - \mathbf{A}] = \begin{bmatrix} z & -1 \\ 2 & z + 3 \end{bmatrix}$$

and

$$|z\mathbf{I} - \mathbf{A}| = z^2 + 3z + 2 = (z + 1)(z + 2)$$

Evaluating the inverse matrix in (2-96) and multiplying by z , we obtain

$$\begin{aligned} z[z\mathbf{I} - \mathbf{A}]^{-1} &= \begin{bmatrix} \frac{z(z+3)}{(z+1)(z+2)} & \frac{z}{(z+1)(z+2)} \\ \frac{-2z}{(z+1)(z+2)} & \frac{z^2}{(z+1)(z+2)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2z}{z+1} + \frac{-z}{z+2} & \frac{z}{z+1} + \frac{-z}{z+2} \\ \frac{-2z}{z+1} + \frac{2z}{z+2} & \frac{-z}{z+1} + \frac{2z}{z+2} \end{bmatrix} = \mathcal{Z}(\Phi(k)) \end{aligned}$$

Thus

$$\underline{\Phi(k) = \mathcal{Z}^{-1}[z[z\mathbf{I} - \mathbf{A}]^{-1}] = \begin{bmatrix} 2(-1)^k - (-2)^k & (-1)^k - (-2)^k \\ -2(-1)^k + 2(-2)^k & -(-1)^k + 2(-2)^k \end{bmatrix}}$$

2.12 Linear Time-Varying Systems

$$\begin{cases} x(k+1) = \underline{A(k)}x(k) + \underline{B(k)}u(k) \\ y(k) = \underline{C(k)}x(k) + \underline{D(k)}u(k) \end{cases} \quad (2.104-105)$$

$$\mathbf{x}(1 + k_0) = \mathbf{A}(k_0)\mathbf{x}(k_0) + \mathbf{B}(k_0)\mathbf{u}(k_0) \quad k_0: \text{Initial time}$$

Thus

$$\begin{aligned} \mathbf{x}(2 + k_0) &= \mathbf{A}(1 + k_0)\mathbf{x}(1 + k_0) + \mathbf{B}(1 + k_0)\mathbf{u}(1 + k_0) \\ &= \mathbf{A}(1 + k_0)[\mathbf{A}(k_0)\mathbf{x}(k_0) + \mathbf{B}(k_0)\mathbf{u}(k_0)] + \mathbf{B}(1 + k_0)\mathbf{u}(1 + k_0) \\ &= \mathbf{A}(1 + k_0)\mathbf{A}(k_0)\mathbf{x}(k_0) + \mathbf{A}(1 + k_0)\mathbf{B}(k_0)\mathbf{u}(k_0) + \mathbf{B}(1 + k_0)\mathbf{u}(1 + k_0) \end{aligned}$$

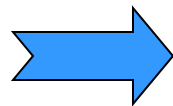
In a like manner, we see that

$$\begin{aligned} \mathbf{x}(3 + k_0) &= \mathbf{A}(2 + k_0)\mathbf{x}(2 + k_0) + \mathbf{B}(2 + k_0)\mathbf{u}(2 + k_0) \\ &= \mathbf{A}(2 + k_0)\mathbf{A}(1 + k_0)\mathbf{A}(k_0)\mathbf{x}(k_0) + \mathbf{A}(2 + k_0)\mathbf{A}(1 + k_0)\mathbf{B}(k_0)\mathbf{u}(k_0) \\ &\quad + \mathbf{A}(2 + k_0)\mathbf{B}(1 + k_0)\mathbf{u}(1 + k_0) + \mathbf{B}(2 + k_0)\mathbf{u}(2 + k_0) \end{aligned}$$

$$\Phi(k, k_0) = A(k-1)A(k-2)\cdots A(k_0) = \begin{cases} \prod_{j=k_0}^{k-1} \underline{A(j)}, & k > k_0 \\ I, & k = k_0 \end{cases} \quad (2.106)$$

$$x(k) = \Phi(k, k_0)x(k_0) + \sum_{j=k_0}^{k-1} \Phi(k, j+1)\underline{B(j)}u(j) \quad (2.107)$$

- See Summary (P77)
- Problems: 2-1, 2-3, 2-6, 2-7, 2-8
2-10, 2-23, 2-26



Test Bank